

B.Sc. in Computer Science and Engineering Thesis

**Performance and Explainability Behavior of Ensemble Machine Learning Models on
Healthcare Imbalanced Datasets**

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CANDIDATES' DECLARATION

This is to certify that the work presented in this thesis, titled, “Performance and Explainability Behavior of Ensemble Machine Learning Models on Healthcare Imbalanced Datasets”, is the outcome of the investigation and research carried out by us under the supervision of Nafisa Tabassum.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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CERTIFICATION

This thesis titled, “**Performance and Explainability Behavior of Ensemble Machine Learning Models on Healthcare Imbalanced Datasets**”, submitted by the group as mentioned below has been accepted as satisfactory in partial fulfillment of the requirements for the degree B.Sc. in Computer Science and Engineering in June 2026.

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ABSTRACT

A large number of healthcare datasets suffer from the problem of class imbalance, which refers to situations where cases belonging to the minority class, which represent a medical issue or illness, happen less frequently than those pertaining to the majority class. Class imbalance is a problem in machine learning and often affects the performance of models, which tend to give priority to the majority class. Thus, such classification may result in the neglect of important minority classes. In healthcare applications, this may be problematic because a wrong prediction might be made. Predictive performance and explanation are thus crucial in such cases. In this paper, I examine the performance and explainability behavior of ensemble machine learning algorithms on imbalanced healthcare datasets. Five datasets, including two breast cancer, one thyroid disorder, chronic kidney disease, maternal health risk, and workplace mental health burnout data, are analyzed in this research. For this purpose, some data preprocessing procedures are implemented, such as cleaning the data and addressing class imbalance using oversampling techniques. Logistic regression, random forest, gradient boosting, k-nearest neighbor, support vector machine, and XGBoost are among machine learning models utilized in this research. Their performance and behavior are examined using Explanation-Guided Ensemble Selection (XGES). Explainability is thus included in the selection procedure for constructing ensembles of classifiers. In this manner, a proper combination of models is selected based on the assessment of performance and explainability. Machine learning algorithms' classification performance is examined using common metrics, such as accuracy, precision, recall, F1-score, and AUC-ROC curves. According to experimental results, the proposed method performs comparably well and even better in some cases than individual machine learning models used in this study. More specifically, the application of explainability to the model selection allows for analyzing prediction behaviors.

Chapter 1

Introduction

Machine learning has found applications in different spheres of life and in medicine, among other sectors. Machine learning can be used for such purposes as diagnosing a disease, identifying possible risks, monitoring the patient and making clinical decisions. With machine learning, medical experts can rely on historical data to make timely and correct conclusions about a person's health condition. However, there is the problem of class imbalance in healthcare data, which means that the dataset is characterized by an unequal number of samples of two classes. Such an imbalance occurs when one class comprises more elements than the other, and instances of the minority class are associated with high-risk individuals or conditions that require prompt actions. As a rule, the presence of class imbalance leads to a deterioration of the machine learning model's performance. Traditional classification models can provide good results in terms of predictive power but cannot detect important examples of the minority class due to a low recall rate. Thus, the problem of class imbalance needs to be solved in order to improve the effectiveness of the algorithm. There exist a variety of approaches to solving this problem; one of the best is considered to be using the ensemble method. The technique of ensemble learning helps to increase a classification system's accuracy, robustness and generalizability. However, finding the right ensemble for a certain task can prove difficult, and in case of healthcare applications, there arise additional difficulties. The main reason is that a model cannot work properly if it is unable to justify its decisions. Nowadays, the domain of Explainable Artificial Intelligence (XAI) helps to resolve this dilemma since XAI methods allow us to gain insight into how machine learning models operate. This research focuses on applying XAI in selecting the optimal ensemble and investigates several ensembles.

1.1 BACKGROUND OF THE STUDY

Machine Learning (ML) has become crucial technology in modern healthcare to assist various procedures including diagnosing diseases, predicting risks, monitoring patients' condition, and making clinical decisions. Analyzing large amounts of health-related data, ML models find patterns and provide insights which contribute to decision-making of healthcare professionals. One of the major issues with using healthcare datasets is a class imbalance, which means that there are considerably more examples of one class than another. For instance, if we consider records of patients suffering from a certain rare disease, we will observe that there are far fewer of them compared to healthy patients. The presence of imbalance affects machine learning models as they usually favor the majority classes and ignore minority ones, resulting in misclassification. [9]

Ensemble learning is a widely used approach in addressing the problem of imbalance. Ensemble method consists in combining several ML algorithms to increase the model's prediction power and its ability to generalize. Various

techniques have been developed to build an efficient ensemble model, such as Gradient Boosting, XGBoost, and Random Forest, among others. The main advantage of such models is their ability to work successfully with healthcare data. At the same time, it has been observed that ensemble approaches are difficult to interpret which hinders healthcare application. [10]

Explainable artificial intelligence (XAI) allows finding appropriate explanations to ML predictions thus increasing its interpretability. Explainability is especially important in applications related to healthcare as they require being trusted and compliant with existing regulations. Therefore, when working with healthcare data, ML models must be evaluated according to not only prediction metrics but also explainability criteria.

The current paper studies performance and explainability behavior of ensemble machine learning algorithms in working with healthcare datasets with imbalanced classes. [11]

1.2 PROBLEM STATEMENT

Machine learning is popular in predicting illnesses, identifying risks, and aiding medical decision-making in the field of healthcare. However, the majority of healthcare data is imbalanced since some classes have a significantly higher number of samples compared to others. Thus, the number of healthy people will be substantially greater than the number of people with a certain illness. As a result, machine learning algorithms develop an inherent bias towards the majority classes and fail to recognize minority class cases.

On the other hand, ensemble machine learning models can improve predictive power and accuracy since several models are being utilized simultaneously. Yet, the choice of the most efficient algorithm varies greatly depending on the type of healthcare data under analysis. Moreover, the majority of ensemble machine learning models can be considered black-box systems whose decision-making mechanisms cannot be easily understood.

In healthcare, accuracy alone does not suffice; thus, doctors require an understanding of a system's predictions. That is why there is a need to develop models that would provide reliable and accurate classifications and be explainable at the same time. However, currently, the vast majority of scientific literature emphasizes the importance of achieving high classification accuracy while paying less attention to explainability.

1.3 SCOPE OF THE STUDY

This research defines the scope of the study to indicate the range of data, methodologies, and technologies involved in this project. It describes what is covered by the project and what is not covered.

Focus on Structured Healthcare Data: This study involves only the analysis of tabular electronic health records. No other types of unstructured medical data (e.g., images such as X-rays, MRIs, clinical text notes, or genomic sequences) are analyzed in the study.

Specific Target Datasets: Five particular binary or multi-class imbalanced medical datasets available at the Kaggle repository are selected: Breast Cancer, Cleaned Thyroid, Kidney Disease, Maternal Health Risk, and Tech Mental Health Burnout.

Machine Learning Techniques: The analysis is limited to the use of conventional machine learning algorithms and ensemble modeling techniques. Base classifiers include Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and XGBoost. No deep learning models (e.g., Neural Networks) are utilized.

Mitigation of Class Imbalance Problem: In order to mitigate class imbalance problem, the data preprocessing step is limited to Synthetic Minority Over-sampling Technique (SMOTE) utilization. More advanced methods or techniques for balancing classes are not considered within this study.

Explainable Artificial Intelligence (XAI) Frameworks: The interpretability analysis is limited to the use of the post-hoc, model-agnostic Explainable Artificial Intelligence frameworks. SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) methods are utilized to analyze global and local explainability.

Operational Environment: The framework is developed and tested in the form of offline simulation using historical benchmark data. No real-time integration into clinical environment or existing hospital Electronic Health Record systems is provided.

1.4 MOTIVATION

The motivation for conducting this research stems from the growing adoption of machine learning applications in the health sector and challenges arising from data imbalance issues. For instance, in applications related to disease diagnosis and risk assessment, the minority cases become crucial for identifying relevant conditions in patients. Machine learning models generally tend to emphasize major classes, leading to underperformance when trying to detect significant minority cases.

Several ensemble learning techniques have been proposed over the years to enhance the accuracy of classification on complicated and data-imbalanced situations. Nonetheless, determining the best ensemble technique among different models is always an issue since each method works uniquely for a certain dataset. In addition, some of the highly performing models may lack interpretability, limiting their use in the medical field.

There has been an increased desire in recent times for reliable healthcare decision support systems that use transparent AI techniques. This motivation comes in handy due to the introduction of XAI in the AI field. Through performance and explainability analysis, this research hopes to achieve a reliable ensemble model that performs optimally and is easy to interpret.

However, traditional machine learning models tend to miss out on those few cases of rare minority classes. Ensemble algorithms enhance the outcome by integrating many models, although identifying the most optimal combination requires time. Explainable Artificial Intelligence (XAI) is mostly employed after the training process for the purpose of understanding the model predictions. The healthcare industry and the financial sector require AI-based systems that are accurate and credible at the same time.

1.5 OBJECTIVES OF THE STUDY

The first aim of this research is to analyze the performance of ensemble machine learning algorithms when applied to class imbalanced healthcare data and to understand their explainability behavior. The imbalanced distribution of classes in healthcare data decreases the efficiency of traditional ML techniques in predicting instances from the minority class. As a result, it becomes crucial to develop approaches capable of demonstrating high accuracy alongside explainability.

It is necessary to analyze and compare the performance of different models, namely machine learning algorithms and ensembles on class imbalanced healthcare data. Moreover, there is a need to explore the explainability behavior using XAI techniques and thus reveal differences between models.

One of the main objectives of the research is to design and test an approach to ensemble selection that relies on explainability information. In particular, the developed strategy will be aimed at facilitating decision-making regarding the selection of suitable ensemble models based on the analysis of performance metrics and explainability behavior.

The overall goal is to select models that demonstrate the highest reliability, accuracy, and explainability when applied to class imbalanced healthcare datasets.

1.6 EXPECTED OUTCOMES

The expected outcomes of this research are as follows:

- **Development of an Explainable Ensemble Learning Framework:** This research is expected to develop an Explanation-Guided Ensemble Selection (XGES) framework that combines predictive performance and Explainable Artificial Intelligence (XAI) to develop a reliable stacking ensemble for imbalanced healthcare datasets.
- **Improved Classification of Imbalanced Healthcare Datasets:** The proposed framework is expected to improve the classification performance of machine learning algorithms on imbalanced healthcare datasets through proper preprocessing, oversampling, and ensemble learning techniques. It aims to enhance the detection of minority-class disease instances while maintaining balanced overall performance.
- **Transparency and Interpretability of Predictions:** By integrating Explainable Artificial Intelligence techniques such as SHAP and LIME, the proposed framework is expected to generate transparent and interpretable predictions that highlight the contribution of important clinical features, thereby increasing users' trust in intelligent healthcare decision-support systems.
- **Comparative Performance Analysis of Machine Learning Models:** The study is expected to compare the performance of Logistic Regression, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost using multiple healthcare datasets. The comparison will be conducted using evaluation metrics including Accuracy, Precision, Recall, F1-score, ROC-AUC, and AUPRC to identify the most suitable models for ensemble construction.
- **Explanation-Guided Model Selection:** The proposed XGES framework is expected to demonstrate that combining predictive performance with explanation stability and explanation diversity leads to more effective model selection than relying solely on predictive accuracy.
- **Contributions to Explainable Healthcare Artificial Intelligence:** The research is expected to contribute to the advancement of Explainable Healthcare AI by providing a practical framework that simultaneously considers predictive performance and model interpretability for trustworthy healthcare applications.

1.7 KNOWLEDGE GAP IDENTIFICATION

Machine learning and ensemble learning have proved to be effective approaches in healthcare prediction. However, the majority of previous studies concentrate more on improving the performance of predictive models rather than ensuring model interpretability and explainability. XAI tools, such as SHAP and LIME, are generally used for post hoc explanation and are not applied in the process of model selection. In addition, existing ensemble selection

approaches do not take into account explanation stability and explanation diversity while selecting base models, especially for healthcare datasets where there is class imbalance. Such an issue generates a knowledge gap for creating an approach that will combine good classification performance and explainability. Therefore, the research suggests an approach called Explanation-Guided Ensemble Selection (XGES), which combines classification performance, explanation stability, and explanation diversity into the process of ensemble selection.

1.8 SUSTAINABLE DEVELOPMENT GOALS (SDGS)



Figure 1.1: United Nations Sustainable Development Goals (SDGs). [1]

Sustainable Development Goals (SDGs) The present research relates to the following United Nations Sustainable Development Goals (SDGs): Goal 3: Good Health and Well-being and Goal 9: Industry, Innovation and Infrastructure. In particular, the major contribution of this research thesis is to SDG 3 as it proposes the Explanation-Guided Ensemble Selection (XGES) framework to enhance disease prediction through the application of healthcare datasets (i.e., Breast Cancer, Thyroid Disease, Kidney Disease, Maternal Health Risk, and Tech Mental Health Burnout datasets). Specifically, the problem addressed in this research thesis relates to the use of oversampling techniques as well as the evaluation of multiple machine learning models to detect minority disease cases due to class imbalance in the healthcare data. Also, Explainable Artificial Intelligence (XAI) techniques (e.g., SHAP, LIME) contribute to the transparency of predictions and improve the overall clinical decision making.

In addition, the present research proposal will make its contribution to SDG 9: Industry, Innovation and Infrastructure by introducing a novel Explanation-Guided Ensemble Selection (XGES) framework. Namely, unlike other ensemble learning approaches that pay attention exclusively to predictive performance, the proposed framework takes into consideration both the explainability and predictive performance of the algorithm in order to select models for an ensemble. Specifically, XGES framework involves Logistic Regression, K-Nearest Neighbors (KNN),

Support Vector Machine (SVM), Random Forest, Gradient Boosting, and XGBoost models within a stacking ensemble while accounting for explanation stability and diversity.

1.9 THE ROLE OF EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) IN HEALTHCARE

Another way of using AI technology in healthcare effectively includes employing Explainable Artificial Intelligence (XAI). Despite the ability of various advanced machine learning methods and ensemble models to predict results accurately, these solutions tend to be quite complex and act like black boxes that make it hard for healthcare professionals to understand the process of decision-making.

In the healthcare sector, decisions might have a serious impact on patients' diagnostics, treatment, and their health state in general. Hence, doctors and other medical workers should be able to trust AI predictions to apply the information obtained from these solutions. XAI helps increase the level of trust in the algorithms used since it enables users to know what influences the decision.

Explainability is important in terms of enabling healthcare professionals to verify the reasonableness of the decisions made by models. Moreover, it can help identify potential issues, such as errors, bias, or discrimination, as well as any imbalanced classes, which are prevalent in datasets in the field of medicine. In particular, minority class cases might refer to rare disorders that can be neglected if not accounted for. [12]

As for Explainable Artificial Intelligence, it plays an important role in the current healthcare sector due to the benefits of machine learning models becoming easier to interpret and more understandable for humans. Although advanced machine learning and ensemble machine learning models provide high-quality results, sometimes they work like black box mechanisms, which makes understanding why particular predictions have been provided challenging.

Decisions based on artificial intelligence algorithms are critical since they influence patients' lives. Inaccuracies and mistakes may cause patients' health conditions to deteriorate. In this regard, healthcare professionals require algorithms providing not only results but also explanations of how they were obtained. XAI allows for getting such explanations and identifying the features that affect the decision-making process.

In addition, XAI can increase the level of transparency and thus help ensure that predictions are correct and reasonable. The algorithm allows clinicians to check whether AI predictions fit into reality and correspond to clinical knowledge. Moreover, the use of XAI contributes to enhancing trust toward artificial intelligence-based solutions because the mechanism behind them becomes obvious.

One more essential aspect that has to be considered when using XAI refers to biases and inaccuracies that can occur while working with certain data. Health-related datasets are often imbalanced, meaning that the number of cases from a minority class is significantly lower than from another class. Such issues may impact machine learning predictions negatively since minority class samples will not be considered properly. Thus, XAI will reveal whether particular cases are ignored by algorithms and what features of those are overlooked.

Thus, incorporating XAI into ensemble machine learning models will improve the latter not only in terms of performance but also in terms of interpretability [11]

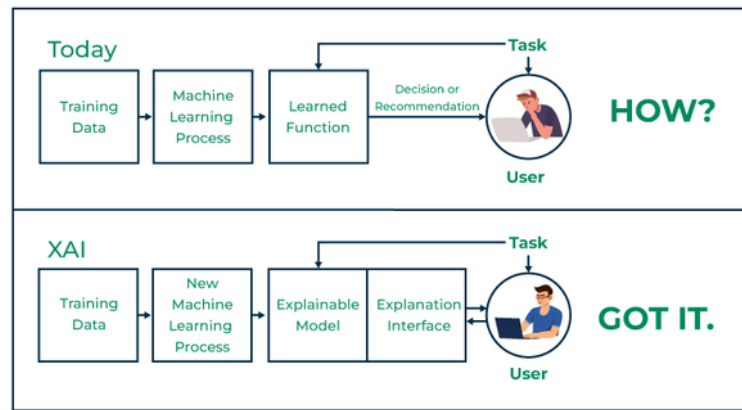


Figure 1.2: XAI [2]

1.10 SUMMARY

This chapter provided background information on the area of interest, including an overview of the challenges faced in the development of machine learning models in the health care domain. The adverse effects of data imbalance on the performance of such models were identified in this section. However, it also noted the ability of ensemble machine learning algorithms to boost the prediction capabilities of such models. Moreover, the problem statement for this thesis was stated to illustrate the need for highly accurate and interpretable machine learning algorithms. The objectives of this project were listed in order to ensure that the evaluation of the models' behavior is achieved. Finally, the role of Explainable Artificial Intelligence in health care was described to emphasize the necessity of using XAI in this field.

Chapter 2

Literature Review

2.1 CONTEXTUAL BACKGROUND

Nowadays, machine learning has become a critical area for healthcare applications, where predictive models are used for disease diagnosis and risk estimation. In practice, the available data are mostly imbalanced, leading to the dominance of majority-class samples and poor predictive performance on minority-class samples by machine learning algorithms. To cope with such limitations, researchers have started investigating sophisticated learning approaches that could enhance classification on imbalanced datasets [13] [3]

The use of ensemble learning in classification tasks has been extensively studied and documented by researchers. Mienye and Sun (2022) provided a review of common ensemble learning approaches such as bagging, boosting, and stacking, emphasizing the superiority of ensemble algorithms compared to single machine learning models. Moreover, ensemble learning helps to enhance classification results [14]

Ensemble learning has been applied successfully in healthcare prediction systems; for instance, Mienye and Sun (2021) investigated approaches aimed at minimizing costs associated with imbalanced medical data, finding that cost-sensitive learning helps detect minority-class instances [13].

In recent years, there has been a noticeable increase in interest in Explainable Artificial Intelligence (XAI), resulting in numerous studies that promote transparency in machine learning models. For instance, Dwivedi et al. (2023) presented core XAI concepts, techniques, and solutions, while Molnar (2019) emphasized the importance of interpretable machine learning methods [2, 11, 15].

Moreover, several recent studies have investigated the integration of ensemble learning and explainability. Jiang et al. (2023) proposed an XAI-based cross-ensemble feature ranking methodology, whereas Bommer et al. (2024) introduced a ranking and evaluation framework based on robustness, faithfulness, and interpretability metrics [16–18].

The Imbalance Problem: Traditional machine learning classifiers tend to be biased toward the majority class in imbalanced healthcare datasets, making the accurate detection of minority-class disease cases more challenging. Consequently, advanced learning algorithms are required to improve minority-class prediction and reduce diagnostic errors [3].

Ensemble Advantage: Advanced ensemble learning techniques, including bagging, boosting, and stacking, generally outperform individual machine learning classifiers by improving predictive performance and model robustness, although they often increase computational complexity [4, 14].

XAI Importance: The integration of post-hoc explainability techniques such as SHAP, LIME, and feature importance analysis is essential for transforming black-box ensemble models into transparent and trustworthy clinical decision-support systems [6, 11, 15].

2.2 EXISTING STUDIES

Machine learning frameworks and ensemble learning architectures have been extensively applied in the healthcare domain for solving various classification problems, especially those characterized by the existence of extreme class imbalances [13, 14].

As noted earlier, clinical datasets tend to have class skewness due to the abundance of healthy subjects relative to the few individuals who are diagnosed with diseases. For the last two decades, an extensive body of literature has demonstrated that regular, independent classifiers cannot handle this problem well and usually produce high classification accuracy but with very low recall and sensitivity to the minority class [3, 5].

Earlier empirical investigations clearly indicate that the use of ensemble learning approaches such as bagging, boosting, and stacking greatly increases the predictive power of the algorithms, whereby the output of several heterogeneous or homogeneous classifiers is combined in order to take advantage of their complementary strengths. The application of such techniques reduces structural variance and localized biases by combining different algorithms [4, 14, 19].

Boosting algorithms (such as AdaBoost and XGBoost) achieve this by progressively minimizing training errors by placing greater emphasis on difficult-to-classify minority instances, while bagging algorithms (such as Random Forest) reduce overfitting through bootstrap aggregation of multiple classifiers. Stacking generalization further improves predictive performance by combining the outputs of base classifiers using a meta-learner [4, 8, 14].

On the other hand, several studies have investigated both data-level and algorithm-level approaches for handling class imbalance. At the data level, heuristic resampling techniques such as SMOTE and ADASYN have been widely adopted to balance class distributions before model training, thereby improving minority-class learning [3].

At the algorithm level, cost-sensitive learning has received considerable attention as an effective strategy for addressing class imbalance. Rather than modifying the class distribution, cost-sensitive methods assign higher misclassification costs to minority-class instances, encouraging classifiers to improve minority-class prediction performance [5].

2.3 CURRENT PRACTICES

Currently, the trends in industry practice and in academia focus on the enhancement of predictive performance of machine learning models when applied to very unbalanced clinical distributions. It is done through the combination of advanced algorithmic fine-tuning, cutting-edge data oversampling techniques and mixed-learning approaches. In modern frameworks, the usage of data-level methods such as the Synthetic Minority Over-sampling Technique (SMOTE) is the standard way to artificially increase the number of synthetic examples of the minority class in order to ensure that the boundaries between classes in the training stage will not be heavily biased toward the majority class.

In addition to the previously mentioned data-level solutions, nowadays, there is a tendency to apply algorithm-level approaches, among which, cost-sensitive learning is one of the most common. Using customized asymmetric loss functions or imposing larger penalties on classification mistakes for the minority class, researchers successfully

manage to overcome the problem of false negatives – the crucial parameter in terms of patient safety in medicine. Moreover, advanced feature selection methods (recursive feature elimination and tree-based methods) are used in modern preprocessing pipelines, which ensures the elimination of noise and redundant parameters and optimization of the dimension of data and isolation of the only important physiological features.

Thus, today, most of the clinical prediction tools and medical decision support applications use the ensemble models as their base. The necessity to use the ensembles is dictated by the nature of ensemble architecture, which allows dealing with highly multidimensional, heterogeneous and non-linear clinical data sets which cannot be adequately modeled using traditional linear or distance-based estimators. By implementing robust ensemble techniques such as random forests, XGBoost and stacked meta-learning generalizations, the contemporary healthcare diagnosis systems achieve the desired stabilization of predictions and decrease of the generalization error.

2.4 ENSEMBLE LEARNING TECHNIQUE

Ensemble learning approach is a state-of-the-art machine learning approach whereby a number of base learners (individual learners) are trained and aggregated for solving a particular computing problem or task. Ensemble learning approach is based on the assumption that a well-planned combination of a large number of weak or fairly strong estimators can help in creating a strong meta-learner that minimizes the generalization error. It has been observed from a large body of academic literature that the ensembles are substantially better than individual algorithmic solutions such as traditional decision tree, logistic regression, or distance-based approaches, especially in case of real-world health care data sets, which are highly affected by environmental noise, feature redundancy, and class imbalances.

Ensemble learning approach helps in managing the bias-variance dilemma. For obtaining the best possible classification boundary, modern ensemble learning algorithms have been classified into three different structural paradigms, i.e., bagging, boosting, and stacking. [13]

2.5 EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

However, as modern machine learning architectures advance, their mathematical decision-making processes become increasingly complex. Although deep neural networks and modern ensemble models achieve remarkable predictive performance, they generally operate as black-box systems whose internal reasoning is difficult to interpret [2, 14]. This lack of transparency creates significant challenges in high-stakes domains such as healthcare, where unreliable or unexplained predictions may adversely affect clinical decision-making and patient safety. To address this limitation, Explainable Artificial Intelligence (XAI) has emerged as an important research field that aims to provide human-understandable explanations for machine learning models while maintaining their predictive performance.

In clinical informatics, model interpretability is closely associated with user trust, diagnostic validation, regulatory compliance, and responsible AI adoption [20, 21]. Without understanding why a particular prediction has been generated, healthcare professionals may hesitate to rely on AI-assisted diagnostic systems. XAI addresses this challenge by providing both global and local interpretability. Global explanations describe the overall behavior and feature importance of a model across the entire dataset, whereas local explanations reveal the contribution of individual features to a specific prediction [17, 22].

2.5.1 SHAP (SHAPLEY ADDITIVE EXPLANATIONS)

SHAP is an analytically well-founded framework based on cooperative game theory, using the concept of Shapley values. In this algorithm, a predictive problem is considered a cooperative game, in which input features are regarded as players and the predictive probability as the payoff. SHAP computes the average marginal contribution of every feature across all possible subsets of features (coalitions) [6, 15].

The first and foremost benefit of the SHAP method lies in its strict adherence to several important mathematical properties [6]:

- **Efficiency (Local Accuracy):** The sum of the SHAP values of all input features equals the difference between the baseline prediction and the prediction for a specific instance.
- **Symmetry:** Two features receive identical SHAP values if they contribute equally across all possible feature coalitions.
- **Consistency (Monotonicity):** If the contribution of a feature increases in an updated predictive model, its corresponding SHAP value will not decrease.

Adherence to these theoretical properties makes the feature attribution process mathematically consistent and globally coherent, allowing healthcare professionals to understand how individual physiological factors, such as blood pressure, age, or tumor size, influence a patient's predicted diagnosis [6, 11, 15].

2.5.2 LIME (LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS)

In contrast to the SHAP method, LIME strictly concentrates on finding locally optimized explanations. While the assumptions of LIME are based on the notion that a global model is too complicated to be explained linearly, the predictions and decision surfaces of such a complex model can be approximated using a simpler and interpretable model, such as linear regression or shallow decision tree in the vicinity of one particular observation only.

The workflow in LIME consists of a few stages described below:

Permutation: The feature values of the targeted patient record are slightly changed to form a cloud of artificially generated data points around the initial record.

Prediction: The "black-box" global model predicts probability scores for all newly created data points.

Weighting: A proximity kernel provides the highest weights to those data points that are closer to the targeted observation.

Optimization: A simple regularized linear model is trained on the weighted dataset of new observations in order to find out local decision boundaries.

Linear coefficients obtained as a result give a clear picture of local factors that were responsible for the given classification of the targeted patient. Combination of SHAP and LIME allows obtaining a two-layered explanation approach for modern healthcare predictive processes.

[12]

2.6 SUMMARY

The analyzed literature shows that ensemble learning approaches have the potential to considerably enhance classification tasks in the case of imbalanced data. In addition, XAI methods have proven to be indispensable when it comes to improving the transparency of AI-driven solutions. While there exist a number of studies dedicated to ensemble learning or explainability individually, there is very little literature available on how both concepts interact in relation to healthcare imbalanced data.

Chapter 3

Materials and Datasets

3.1 DATASET INFORMATION

In order to assess the predictive performance and explainability properties of the proposed Explanation-Guided Ensemble Selection (XGES) approach properly, we use five different healthcare-related benchmark datasets sourced from Kaggle database. The selected benchmarks include Breast Cancer, Thyroid Diseases, Kidney Diseases, Maternal Health Risk and Tech Mental Health Burnout. By selecting these particular datasets, we create an inherently heterogeneous testing environment since they represent various branches of medicine, including critical oncology and endocrinology disorders, as well as mental health measurements and maternal physiology.

One of the key structural characteristics of all five selected datasets is their systematic imbalance. It is one of the common problems that appear in clinical informatics as the number of positive examples of the illness (minority class) is considerably higher than the number of negative examples (majority class). From the perspective of a clinical environment, this kind of imbalance is natural, since the majority of people in the general population is healthy. However, computationally, it brings a lot of issues as machine learning algorithms tend to optimize global empirical loss function and minimize it by choosing the majority class, which may cause fatal errors. [23]

3.1.1 BREAST CANCER DATASET

Breast Cancer Dataset represents a binary classification problem used for predicting whether the detected tumor is benign or malignant, with most samples belonging to the former class and the latter one constituting the minority class.1. Breast cancer.csv Size: 569 rows and 33 columnsTarget Variable: DiagnosisBenign (B): 357 samples-Malignant (M): 212 samplesNull Values: None, except for an empty last column titled "Unnamed: 32" with 569 null values that need to be excluded from the data set during pre-processing Key Attributes: patient id and diagnosis.Ten numeric attributes, calculated for each nucleus from digitized image FNA breast mass: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

3.1.2 THYROID DISEASE DATASET

Thyroid Disease Dataset is used for determining whether the patient has any thyroid disorder by using several health-related attributes. This dataset possesses class imbalance between healthy patients and those having a disease. Thyroid Dataset with Cleaning (cleaned dataset Thyroid1.csv)Size: 3,771 rows and 26 columnsLabel: bina-

ryClassNegative / Normal (0): 3,480 casesPositive / Thyroid Diseased (1): 291 casesNumber of Missing Cases: 0 missing values (this data set is complete).Features:Demographic Variables: Age, Gender.Disease History/Presence of Symptoms (binary variables): On thyroxine, Query on thyroxine, On anti-thyroid medication, Sick, Pregnant, Thyroid surgery, I131 therapy, Query Hypo-thyroid, Query Hyperthyroid, Lithium, Goitre, Tumor, Hypopituitary, Psych.Laboratory Results / Indicator Variables: Quantitative scores for various hormones such as TSH,

3.1.3 KIDNEY DISEASE DATASET

Kidney Disease Dataset can be used to predict chronic kidney diseases through analyzing different numerical and categorical health attributes. It often serves as an object of analysis for healthcare-related problems. Kidney Disease Data Set (kidney disease.csv)Shape: 400 rows, 26 columnsTarget Variable: classificationChronic Kidney Disease (ckd): 248 samplesNot Chronic Kidney Disease (notckd): 150 samplesImportant: There are 2 samples with formatting as ckdwhich have an extra tab at the end and need to be cleaned for text.Missing Values: Have many missing values in several columns, especially:rbc (Red Blood Cells): 152 missing samplesrc (Red Blood Cell Count): 130 missing sampleswc (White Blood Cell Count): 105 missing samplespot (Potassium) sod (Sodium): 88 and 87 missing samples respectively Key Attributes: Clinical measurements like blood pressure (bp), specific gravity (sg), albumin (al), sugar (su), random blood glucose (bgr), blood urea (bu), serum creatinine (sc), and hemoglobin (hemo).

Classification risk factors: high blood pressure (htn), diabetes mellitus (dm), coronary artery disease (cad), appetite condition (appet), pedal edema (pe), and anemia (ane).

3.1.4 MATERNAL HEALTH RISK DATASET

Maternal Health Risk Dataset assesses health risks to which pregnant women can become exposed depending on their age, blood pressure, and blood sugar. Different health risk classes are considered. Structure: 1014 rows, 7 columnsTarget Variable: RiskLevelLow risk: 406 recordsMedium risk: 336 recordsHigh risk: 272 recordsNumber of Missing Values: 0Missing Value Ratio: 0 percent Main Variables: A purposeful set of variables capturing crucial health parameters of the mother: Age, SystolicBP (systolic blood pressure), DiastolicBP (diastolic blood pressure),

3.1.5 MENTAL HEALTH BURNOUT DATASET

Tech Mental Health Burnout Dataset predicts mental health burnout of individuals working in the field of IT. This dataset has features related to gender and race, working place attributes, and mental health issues. Structure: 1014 rows, 7 columnsTarget Variable: RiskLevelLow risk: 406 recordsMedium risk: 336 recordsHigh risk: 272 recordsNumber of Missing Values: 0Missing Value Ratio: 0 present Main Variables: A purposeful set of variables capturing crucial health parameters of the mother: Age, SystolicBP (systolic blood pressure), DiastolicBP (diastolic blood pressure),

This variety of health problems enables researchers to explore the potential of machine learning algorithms to solve healthcare-related issues.Tech Mental Health and Burnout Dataset (tech mental health burnout.csv) Shape: 150,000 Rows, 25 Columns Target Variable: burnout level (categorical) or burnout score (continuous) Low: 131,620 Samples Moderate: 18,315 Samples High: 65 Samples Can alternatively use seeks professional help (0, 1) as a classifier. Missing Data: No missing data. Key Variables:Demographics and Job Role: Age, Gender, Job Role, Years of Experience, Company Size, and Work Mode (Remote, Hybrid, Onsite). Workplace Stressors: Hours Worked per Week,

3.2 CURRENT PRACTICES

The data preparation stage is one of the most critical phases in any machine learning project because the quality of the input data directly affects the performance, reliability, and generalization ability of the developed models. Healthcare datasets often contain missing values, noisy observations, duplicate records, inconsistent formatting, and class imbalance, all of which can significantly reduce model accuracy if they are not handled properly. Therefore, each dataset used in this research underwent a comprehensive preprocessing pipeline before being used for model training and evaluation.

The first step involved examining the datasets for missing values, duplicate records, and inconsistent data entries. Missing values were carefully handled using appropriate preprocessing techniques depending on the nature of the feature. Numerical attributes were imputed using statistical measures where appropriate, while irrelevant or incomplete records that could negatively affect model performance were removed. Duplicate records were identified and eliminated to avoid introducing bias during model training. In addition, inconsistent data formats, typographical errors, and invalid entries were corrected to ensure data integrity and consistency across all datasets.

Since several healthcare datasets contained categorical variables, these attributes could not be processed directly by machine learning algorithms. Therefore, categorical features were transformed into numerical representations using suitable encoding techniques such as Label Encoding and One-Hot Encoding. This transformation enabled machine learning models to interpret categorical information while preserving meaningful relationships among the variables.

Feature scaling was applied whenever necessary to normalize the range of numerical features. Many machine learning algorithms, particularly Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), are highly sensitive to differences in feature magnitude. Without proper scaling, variables with larger numerical ranges may dominate the learning process and negatively influence the model's predictions. Standardization or normalization techniques were therefore employed to ensure that all numerical features contributed equally during training, resulting in faster convergence and improved model stability. Feature scaling was utilized wherever needed for normalizing the numerical features range. Various machine learning algorithms, including SVM and K-NN, are highly sensitive to the magnitudes of features. In case of absence of adequate scaling, features having a larger numerical range would dominate the learning process and adversely affect the predictions made by the model. Thus, standardization or normalization procedures were performed in order to make sure that all features had an equal contribution towards the learning process, ensuring quick convergence and increased stability of the model.

Another important step in preprocessing consisted of identifying possible outliers and assessing the data distribution in general. Extremely large values could be caused by various factors, including measurement errors, data input mistakes or rarity of certain observations. These observations are likely to have an adverse effect on the learning process. Therefore, statistical methods and visualization techniques were utilized in order to assess the distributions of essential features and detect the outlier observations. Preprocessing steps aimed at decreasing the impact of these outliers while retaining all relevant information from a clinical perspective were chosen accordingly.

One of the most challenging issues in the analyzed healthcare datasets was the problem of class imbalance. Namely, the number of samples belonging to minority classes was significantly lower than that of majority classes. This can cause bias of machine learning models in favor of majority classes. To avoid such limitations, the oversampling techniques were used exclusively on the training dataset. Various oversampling methods were investigated, namely Synthetic Minority Over-sampling Technique (SMOTE), Adaptive Synthetic Sampling (ADASYN) and Borderline-SMOTE.

After performing all the preprocessing steps, the data for each experiment was split into two parts: the training and testing sets through stratified sampling. Stratified sampling splits the data into two parts such that the distribution of classes is maintained in both the sets. The training set was used to create the machine learning models, while the testing set was kept out-of-bag to assess the effectiveness of the Explanation-Guided Ensemble Selection (XGES) algorithm.

3.3 MACHINE LEARNING MODELS IMPLEMENTED

A number of supervised machine learning models will be used in this research to assess their effectiveness in imbalanced data classification for health care datasets. The models chosen for assessment are Logistic Regression (LR), K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB), and Extreme Gradient Boosting (XGBoost, or XGB). These models have been selected due to the difference in their paradigms of machine learning and high efficiency in solving prediction tasks in the field of health care. Thus, by assessing a range of classification algorithms, it is possible to compare their effectiveness and robustness in various conditions.

The models chosen for assessment include classical machine learning algorithms and ensemble learning algorithms. The former allow using simple and interpretable decision rules, while the latter unite several learners to enhance accuracy and minimize the limitations associated with the use of one classifier. Given that health care datasets may contain complicated dependencies and noisy observations as well as class imbalance problem, the selection of various approaches for learning is necessary. [15]

3.3.1 Logistic Regression (LR)

The Logistic Regression algorithm has been chosen as one of the baseline classifiers due to its simplicity, efficient computations, and interpretability. In contrast to linear regression, the Logistic Regression model makes predictions for the probability of belonging to a certain class utilizing the sigmoid function.

One of the main strengths of Logistic Regression is the interpretability of feature contributions by learning the coefficients of features. Therefore, the logistic regression method works well in areas such as healthcare, where understanding the impact of clinical parameters is crucial. It provides accurate results on linearly separable data sets and serves as a good benchmark for other more complicated methods. On the other hand, the performance of the classifier drops in case of highly nonlinear relations or interactions between features.

3.3.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is an instance-based, non-parametric machine learning algorithm that uses the classes of the nearby data samples to classify new observations in the feature space. The algorithm finds the distance from the new observation to all training data points through a distance measurement like Euclidean distance and assigns the most common class of the neighbors.

The implementation of KNN is easy and requires no training step, making it suitable for structured datasets. Nevertheless, the algorithm depends on the number of Ks used, the selection of distance measure and feature scaling. Because distances are dependent on the size of features, normalization or standardization should be performed before using KNN. With healthcare data, KNN is able to detect patients with the same traits but can be inefficient in big datasets.

3.3.3 Support Vector Machine (SVM)

Support Vector Machine is an advanced supervised machine learning technique used for creating an optimal hyperplane that can increase the distance between different classes. The Support Vector Machine algorithm determines the hyperplane with maximum margin between different classes, thus increasing its generalization power on new examples.

The major advantage of SVM is the ability to work with multidimensional data and perform non-linear classification with the help of the kernel trick, including the Radial Basis Function (RBF) kernel. It makes SVM applicable for many healthcare cases where a large amount of clinical data is used. However, SVM needs appropriate hyperparameter and feature tuning for proper operation. The training process may also be computationally expensive for very large amounts of data.

3.3.4 Random Forest (RF)

The Random Forest classifier is an ensemble method based on the bagging algorithm. It creates multiple decision trees by randomly selecting samples and input variables from the training set. When making predictions, each tree predicts a certain class independently, and the result is obtained via majority voting.

The Random Forest classifier is highly popular due to the following qualities: reliability, high tolerance to overfitting, and the capability to model complex non-linear correlations. Also, this classifier allows estimating the importance of features, which is useful when conducting studies related to the healthcare industry and trying to discover significant clinical factors. Besides that, Random Forest works well with noisy and incomplete data sets, so it is one of the most reliable classifiers for medical applications.

3.3.5 Gradient Boosting (GB)

Gradient Boosting is an ensemble learning method which involves training a series of weak learners, usually decision trees, in order to reduce the prediction errors. In contrast to bagging techniques, each subsequent tree tries to learn from the errors of previous trees using gradient descent optimization.

The above learning procedure allows Gradient Boosting to capture complicated data relationships and produce accurate predictions. The method works especially well on data where there is complex interaction between the features, which is quite common in healthcare data sets. On the other hand, Gradient Boosting is much more computationally expensive compared to Random Forest.

3.3.6 Extreme Gradient Boosting (XGBoost)

The Extreme Gradient Boosting algorithm is an optimized implementation of the Gradient Boosting method meant to increase computational efficiency and model performance. It includes many additional components, such as regularization mechanisms, second-order gradients optimization, tree pruning, parallel computations, and dealing with missing values.

All the above-listed components help make XGBoost capable of generating highly accurate classification models while lowering the chances of overfitting. Due to its excellent results in working with structured tabular data, the Extreme Gradient Boosting algorithm became one of the most popular algorithms for predicting outcomes and diagnosing diseases in medicine.

3.3.7 Evaluation of Predictive Performance of the Models

First of all, after the pre-processing of the data, all six machine learning models were trained and evaluated using the same experimental setting to make sure that the comparison of their performances would be conducted fairly. In terms of prediction accuracy, metrics such as Accuracy, Precision, Recall, F1-score, ROC-AUC, and Area Under the Precision-Recall Curve (AUPRC) were used. As the data was imbalanced, extra focus was paid to the latter three metrics since they are more capable of showing the model's ability to recognize minority-class samples.

Besides that, techniques from Explainable Artificial Intelligence (XAI), specifically SHAP, were applied to study how each model made its predictions. As part of the analysis of the explainability, the importance of features, stability, and diversity of explanations were considered. This allowed for making a thorough comparison of the models not only in terms of prediction accuracy but also of their interpretability. Thus, the obtained scores served as the starting point for developing the proposed Explanation-Guided Ensemble Selection (XGES) approach. The best models were selected to build the stacking ensemble.

3.4 DESIGNING THE ENSEMBLE MODEL

To improve classification performance on healthcare imbalanced datasets, an Explanation-Guided Ensemble Selection (XGES) framework was developed in this study. The framework combines the strengths of multiple machine learning models while incorporating explainability information into the model selection process. The main objective of XGES is to identify ensemble combinations that provide both high predictive performance and meaningful explanations.

Initially, all candidate machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, and XGBoost, were trained individually on the prepared datasets. Their predictive performance was evaluated using appropriate classification metrics. In addition to performance evaluation, explainability analyses were conducted to understand the contribution of different features and the decision-making behavior of each model.

Based on both performance and explainability results, the candidate models were ranked and selected for ensemble construction. Models that demonstrated strong predictive capability and consistent explanation patterns were prioritized. This approach ensures that the final ensemble model is not only accurate but also transparent and reliable.

The selected models were then combined using a stacking-based ensemble technique. In this approach, the predictions generated by the base models serve as inputs to a meta-learner, which produces the final prediction. By leveraging the strengths of multiple classifiers, the ensemble model reduces the weaknesses of individual models and improves overall generalization performance.

The proposed XGES framework aims to create a balanced solution that addresses the challenges of healthcare imbalanced datasets while maintaining model interpretability. This integration of ensemble learning and explainability distinguishes the proposed approach from traditional ensemble methods that focus solely on predictive accuracy. [14]

3.5 XAI APPROACH AND MODEL EVALUATION

In the current study, methods of Explainable Artificial Intelligence (XAI) were utilized in order to analyze the behavior of decisions made by machine learning models and ensembles. With the help of XAI, it became possible to reveal the importance of certain features for model's prediction and to evaluate the degree of their transparency and interpretation. Information received from explanations helped to learn more about the way models performed classification of healthcare data and choose appropriate models for XGES methodology application. Besides, model performance was measured using such metrics as Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score. These metrics served as a means for assessing models' performance at working with imbalanced healthcare datasets and predicting instances belonging to the minority class. [24]

3.6 SUMMARY

Explanation of the methodology followed in this work. A total of five datasets relevant to healthcare were chosen and preprocessed using various approaches like data cleaning, feature encoding, normalization, and class balancing. Multiple machine learning models like Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, and XGBoost were developed and assessed. The Explanation-Guided Ensemble Selection (XGES) approach was adopted to develop an ensemble model by focusing on both accuracy and interpretability. Explainable Artificial Intelligence (XAI) methods were used to interpret model behavior and understand the importance of features. Evaluation criteria such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC were used to evaluate the models.

Chapter 4

Methodology

Explanation-Guided Ensemble Selection (XGES) implementation included multiple systematic procedures. At first, the healthcare datasets were gathered and split into the training and test sets according to the stratified sampling technique. To solve the issue of class imbalance, oversampling methods were employed in relation to the former set, followed by feature scaling.

Then, several machine learning algorithms were used to train various classifiers, such as Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, and XGBoost, based on the prepared training data. Further, after the evaluation of the obtained algorithms on the test set, their performance was measured and recorded.

Afterward, the sample of the training data was created while preserving the initial class distribution. The Explainability Analysis was executed to assess the contribution of particular features to the predictions generated by the models. Subsequently, the model explanation consistency was analyzed by repeating this process multiple times to ensure explanation stability.

Finally, explanation behaviors of different algorithms were contrasted to measure their diversity. More specifically, models which exhibited complementary explanation patterns were determined to be the appropriate candidates for building an ensemble model. Based on the above criteria, each classifier was awarded an aggregated score and ranked in accordance with it.

Further, the top-ranked classifiers were utilized to construct an ensemble model by employing the predictions they made in a stacking scheme, which involved the creation of a meta-learner using them. This process resulted in obtaining the final version of an ensemble model, which was assessed on the test data.

Five imbalanced healthcare datasets were obtained for experimental analysis. The datasets were preprocessed using cleansing, feature processing, and class balancing procedures. Stratified sampling technique was utilized to split data into training and test subsets. Data oversampling procedure was used to enhance the representation of the minority class. Six machine learning models were built on the balanced dataset. The following models were chosen as candidates: Logistic Regression, Random Forest, Gradient Boosting, KNN, SVM, and XGBoost. The performance of each classifier was analyzed based on performance measures. Explaining each of the models was carried out to evaluate the prediction behavior. Information on feature importance was obtained for each machine learning model. The explanation consistency was analyzed to assess explanation stability. Explanation patterns produced by different models were compared. Diversity of explanations generated by models was analyzed. The performance, stability, and diversity measures were combined. Models were ranked based on their overall performance. Top-ranked models were used to construct the ensemble. Base models were combined using an ensemble

learning technique. The predictions of the base models were used to train the meta-learner. The performance of the ensemble was analyzed on the test set. Comparison was made with the performance of classifiers. The proposed framework achieved better performance and explainability. It can be concluded that explainability enhances ensemble building. The constructed ensemble generates reliable predictions on health imbalanced datasets. The Proposed steps, Explanation Guided Ensemble Selection (XGES) optimize the stacked ensemble model in terms of performance, explanation stability, and explanation diversity. The first step includes preparing the data, loading the dataset D , dividing it into training and testing data, balancing the training dataset through SMOTE algorithm and normalizing the features. Then the models, belonging to the set M , are trained on the balanced dataset. Next, their performance is evaluated on the test dataset by calculating Area Under the Precision-Recall Curve (AUPRC) scores. In order to include the explainability, a stratified subset S is extracted from the training data, and SHAP is applied to calculate feature importance vectors for each model.

In the next step, two explainability metrics are calculated. First, the stability metric of the model is estimated by doing k -fold analysis of the subset S and calculating variance of feature importances across the folds. Second, the diversity metric is measured by calculating the similarity of explanation vectors between each pair of models and transforming it to the diversity score. Based on these two metrics, each model is ranked according to the following formula:

$$\text{Score}(m) = a \times \text{Performance} + \beta \times \text{Stability} + \gamma \times \text{Diversity}$$

[25] [26]

Top- k ranked models are selected as base models and further utilized in the stacking procedure where the models are trained to predict meta-features, which serve as input for the meta-learner. Then the stacked ensemble E is evaluated on the test data and returned.

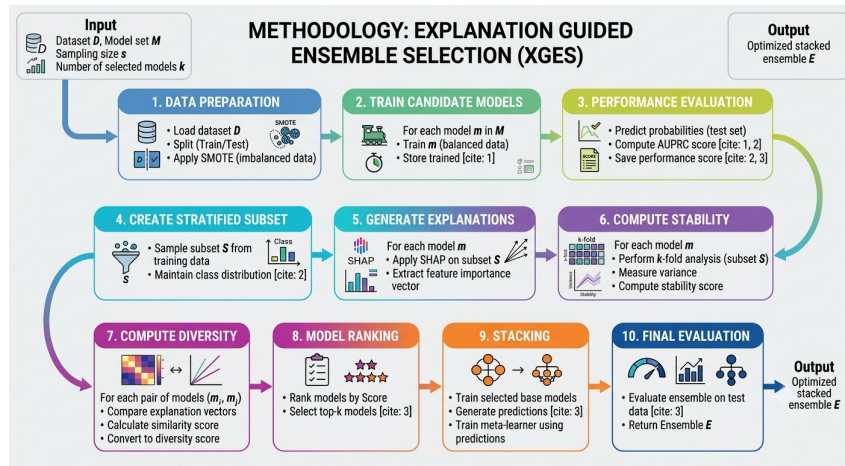


Figure 4.1: Methodology [3] [4] [5] [6] [7] [8]

4.1 THE SCORING FUNCTION

The scoring algorithm rates candidates based on a combination of three critical factors in one score: predictability, explanation stability, and diversity. The algorithm follows the budget principle, wherein the sum of all three parameters is equal to 1. The mathematical constraint enables normalization of scores within 0 and 1 intervals.

Firstly, the performance is measured in terms of metrics resilient to class imbalance problems, which makes the algorithm capable of capturing rare minority cases. Secondly, the explanation stability is assessed using SHAP feature attribution maps in order to rule out models depending on any unreliable short-term tricks. Lastly, the diversity is calculated as a requirement that will force the selection process to pick the models applying different logical reasoning to make decisions.

Through changing weights, the customization options of the selection pipeline become infinite and do not affect scale constraints. A significant value of the stability parameter is optimal for clinical use since it emphasizes reasoning profile verifiable by a doctor. In turn, focusing on diversity yields an ensemble capable of compensating each other's weak spots due to having completely different perspectives.

$$\text{Score} = \alpha \times \text{Performance} + \beta \times \text{Stability} + \gamma \times \text{Diversity} \quad (4.2)$$

$$\text{subject to: } \alpha + \beta + \gamma = 1 \quad \text{where } \alpha, \beta, \gamma \geq 0$$

In this case, (Alpha): Weight assigned to indicate the influence of model performance on the final score.

(Beta): Weight assigned to indicate the influence of explanation stability on the final score.

(Gamma): Weight assigned to indicate the influence of model diversity on the final score.

Performance: Model's predictive accuracy.

Stability: Model's explanation reliability.

Diversity: Model's explanation uniqueness.

[\[11\]](#) [\[14\]](#) [\[6\]](#) [\[27\]](#) [\[7\]](#)

4.2 EVALUATING METRICS

Standard evaluating metrics lack reliability if employed on imbalanced validation distribution samples. Thus, applying the same approach in case of a dataset featuring an IR ratio of 99 : 1, a model with no prior knowledge of the minority class will have 99% Accuracy but no capability to recognize the patients at risk of dying prematurely. In this case, different approaches must be used as follows:

Precision: Represents the percentage of correct positive predictions compared to all positive predictions.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (4.3)$$

Recall (Sensitivity / True Positive Rate): Reflects the model's ability to detect all positive cases.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4.4)$$

F1-Score: The harmonic mean of Precision and Recall.

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.5)$$

Area Under the Precision-Recall Curve (AUPRC): In contrast to Receiver Operating Characteristic Area Under the Curve (AUROC) that may remain quite good because of plenty True Negatives, this measure focuses only on the minority class prediction accuracy.

4.3 OVERSAMPLING TECHNIQUES

Sampling methods are employed by researchers to rectify any potential distortion that may occur at the decision boundary. [9]

A. Random Oversampling (ROS): ROS creates a copy of random samples of the minority class to bring about equalization in the training set. Although computationally cheap, it tends to distort the distribution of the minority class, leading to overfitting to these replicated samples.

B. Synthetic Minority Over-sampling Technique (SMOTE): SMOTE avoids duplication by generating synthetic samples on the line segments between existing samples in the minority class. For any sample x_i from the minority class, its k nearest neighbors are found in the minority class. Next, one of its neighbors x_{nn} is randomly chosen, and a new point is created using:

$$x_{\text{new}} = x_i + \lambda(x_{nn} - x_i) \quad (4.6)$$

where λ is a random uniform variable ranging from 0 to 1 ($\lambda \in [0, 1]$).

4.4 ADAPTIVE SYNTHETIC SAMPLING AND BORDERLINE-SMOTE

Both Adaptive Synthetic Sampling (ADASYN) and Borderline-SMOTE can be considered improved versions of the conventional SMOTE algorithm, whereby both are designed to generate samples based on the topology of the local neighborhood around the minority class. In particular, ADASYN works by generating a density distribution whereby samples from the minority class with a high percentage of majority class samples surrounding them get assigned high sampling weights. On the other hand, Borderline-SMOTE algorithmically separates the minority samples into three categories, namely "Safe," "Danger," and "Noise." The process then generates new samples only from those classified as "Danger" since they are the points of overlap between the minority and majority classes. [9]

4.5 BOOSTING

Boosting is an ensemble learning algorithm that aims at combining several weak learners to develop one strong predictor. In contrast to bagging, which trains several models simultaneously, boosting trains models successively, and the next model tries to fix the prediction error of the previous model. Hence, the performance of boosting improves with each successive step in the sequence.

The main aim of boosting is to decrease both bias and variance of the model and increase its generalization capabilities. Because of the nature of the boosting algorithms, when each learner learns from the mistakes of the previous learners, the accuracy of the final ensemble becomes much higher compared to accuracy of each individual learner. Moreover, boosting algorithms perform better with structured tabular data with complicated relationships among features and thus, they are frequently used in healthcare, finance, fraud detection, and diagnoses of diseases.

One of the strengths of boosting algorithms is that it can model complicated relationships and interactions among features without performing complicated feature engineering. On the other hand, boosting algorithms are sensitive to the choice of hyperparameters. The addition of too many learners and a high value of the learning rate lead to overfitting of the algorithm. Despite the fact that boosting is more computationally intensive compared to bagging, it has better accuracy of predictions.

4.6 STACKED GENERALIZATION

Stacking employs a heterogeneous architecture wherein different families of models that are totally separate from each other (such as Support Vector Machine, Deep Neural Network, Random Forest) are used as estimators at the base level. [14]

Stacking takes place in two levels of operation:

Level-0 (Base Estimators): Different types of candidate estimators are trained using the main dataset. The output of every model is a vector containing class probabilities or continuous estimates.

Level-1 (Meta-Estimator): This meta-estimator (a regularized linear model such as Ridge Regression or Logistic Regression) uses the aggregate predictions from the Level-0 models as inputs to learn the best combination of weights for making predictions.

4.7 EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

With the shift of Artificial Intelligence (AI) towards high-stake clinical applications, post-hoc interpretability tools have become essential for understanding the reasoning behind complex machine learning models. Explainable Artificial Intelligence (XAI) provides techniques that help researchers and clinicians interpret model predictions, thereby increasing transparency, trust, and reliability in healthcare applications.

Global Explainability vs. Local Explainability **Global Explainability** aims to explain the overall behavior of a machine learning model by identifying how input features influence predictions across the entire dataset.

Local Explainability focuses on explaining a single prediction by identifying which features contributed to the prediction for an individual instance.

Permutation Feature Importance Permutation Feature Importance is a global explainability technique that measures feature importance by randomly shuffling one feature at a time and observing the resulting decrease in model performance. A larger decrease indicates that the feature is more important for prediction.

LIME (Local Interpretable Model-agnostic Explanations) LIME approximates the complex decision boundary of a machine learning model around a particular prediction using a simple interpretable surrogate model, such as a sparse linear regression model.

SHAP (SHapley Additive exPlanations) SHAP is a game-theoretic explainability method that assigns each feature a Shapley value representing its contribution to the final prediction. For a given instance, the prediction can be expressed as

$$g(z') = \phi_0 + \sum_{i=1}^M \phi_i z'_i \quad (4.7)$$

where

- ϕ_0 is the base value (average prediction of the model),
- ϕ_i is the Shapley value of feature i ,
- z'_i is the simplified representation of feature i , and
- M is the total number of input features.

SHAP satisfies three important mathematical properties:

- **Local Accuracy (Efficiency):** The sum of all SHAP values equals the difference between the model prediction and the expected prediction.
- **Missingness:** If a feature is missing or has no contribution, its SHAP value becomes zero.
- **Consistency:** If a feature contributes more in an updated model, its SHAP value will never decrease.

4.8 XAI IN HEALTHCARE

In the clinical setting, explainability is not only a matter of analysis; it becomes a structural necessity on several fronts: Operational Ethics and Clinical Trust. Doctors act based on the principle of *primum non nocere* – first, do no harm. A naked probability estimate with Ninety-four percent. probability of malignancy for an asymptomatic patient is an action impossible to take without an explanation of the reasons behind it. With the help of XAI, which explains why some particular variables contributed to making a prediction (for instance, biomarker elevation in combination with familial history), one can correlate this prediction with known pathophysiology and make sure of the soundness of the system’s reasoning prior to taking any action. Regulatory Requirements Clinical software is subject to stringent regulatory control by international authorities. Laws like the right-to-explanation part of GDPR make it mandatory to provide a right to explanations to the decisions made by automated algorithms about individuals. FDA regulations regarding the development of Software as a Medical Device stipulate the requirement of having validation trails and logic paths in order not to let unexplainable bias become a part of treatment. Decision Support ML models serve as a layer of defensive decision support, rather than replace the clinician themselves. XAI points out borderline cases where a tired radiologist could overlook some anomaly or uncovers potential drug interactions in EHR data. Explainable AI was incorporated into the Explanation-Guided Ensemble Selection (XGES) framework, in order to enhance not only the performance, but also the interpretability of the machine learning models. In particular, the Shapley Additive exPlanations (SHAP) method was used to assess the impact of each feature on the machine learning models’ predictions, as well as to compare the stability of explanations between different machine learning algorithms and to study the diversity of the explanations generated by those. [24]

4.9 SUMMARY

This research presented an Explanation-Guided Ensemble Selection (XGES) framework in order to increase the accuracy and interpretability of machine learning models on imbalanced healthcare datasets. At the initial step, the datasets were preprocessed in terms of missing value treatment, encoding categorical variables, feature scaling if needed, and dataset balancing through SMOTE technique. The preprocessed datasets were further split into training and test sets for developing the models. In order to develop six machine learning models (Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, XGBoost), their accuracy was measured using classification performance metrics. In order to increase the interpretability of these models, SHAP technique was used to determine feature importance and provide an explanation for each model’s decision making process. Based on the diversity and stability of explanations, along with accuracy of the models, the ranking was done and the best performing models were chosen and formed an ensemble model by using stacking technique with meta-learner. As a result, the proposed framework was tested on five healthcare datasets based on classification metrics, including Accuracy, Precision, Recall, F1-score, ROC-AUC, and AUPRC. Experimental results showed that inclusion of explainability in ensemble model selection resulted in accurate and interpretable framework for predicting healthcare datasets

Chapter 5

Development and Implementation

5.1 PROPOSED MODEL ARCHITECTURE

The architecture of the proposed solution relies on an Explanation-Guided Ensemble Selection (XGES) framework aimed at enhancing the predictive performance and increasing the degree of interpretability of models on healthcare datasets. Specifically, the presented approach combines the use of multiple machine learning models, explainable artificial intelligence (XAI) solutions, and the stacking-based ensemble method.

The general scheme of the process starts with the collection and pre-processing of healthcare datasets. At this step, all missing values are identified, categorical variables are encoded, and the problem of class imbalance is managed through oversampling techniques. The data samples obtained are then split into training and testing datasets that are used to build predictive models.

Afterwards, multiple machine learning models such as Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, and XGBoost are applied to obtain predictions and assess the performance of models according to multiple metrics. In addition to performance assessment, an analyze of explainability is performed to determine how the contribution of specific features influences the results.

The use of the XGES framework involves the ranking of ml models on the basis of their predictive performance and ability to provide explainability information. Specifically, models with high classification capabilities and the consistency of explanation patterns are chosen as potential components of the ensemble. [17]

5.2 BASE MODEL DEVELOPMENT

Base ml model development represents an essential phase in the development of the proposed XGES framework. In the present study, a variety of machine learning algorithms were chosen and used to explore their performance when applied to the healthcare datasets under consideration. Specifically, the following ML models were used: LR, RF, GB, KNN, SVM, and XGB. These models were chosen due to their differences in learning methods and applicability in health care prediction tasks.

All the chosen models were trained using the preprocessed datasets where missing values were handled, feature scaling performed, categorical variables encoded, and class imbalance solved. Hyperparameters were tuned appropriately to guarantee better performances. As a result, all models learned features from data that helped predict target outcomes. For instance, Logistic Regression was applied as a simple and easily interpretable model. KNN

was chosen due to its capacity to make classifications based on similarities in datasets.

SVM was applied due to its capability to build effective optimal decision surfaces. At the same time, RF and GB were used due to their strong ensemble nature and resistance to overfitting problems. Finally, XGBoost algorithm was used due to its advanced boosting technology and high performance. All these algorithms and models were trained on datasets to learn features and make predictions about the target outcomes.

Subsequently, the performance of the models was measured through the analysis of multiple classification metrics. This analysis allowed for determining the strengths and weaknesses of each model. All trained models can be seen as base models and used as a starting point in ensemble generation within XGES framework. [15]

5.3 CREATION OF THE ENSEMBLE MODEL AND UTILIZATION OF XAI TECHNIQUES

The newly designed Explanation-Guided Ensemble Selection (XGES) paradigm uses the principles of both ensemble learning and XAI to enhance the efficiency of the prediction algorithm and its interpretable nature. At first, various models such as logistic regression, random forest, gradient boosting, k-nearest neighbors, support vector machine, and xgboost were developed and tested separately. The efficiency of each model was measured via classification metrics. In addition, several XAI methods were applied to evaluate the behavior of models by analyzing feature importance and overall decision making. The output of this analysis showed which specific features were used by each particular model to make its prediction and how reliable this behavior was. Based on the above criteria, the most relevant models were chosen for the further ensemble building. The chosen models were combined together through the use of a stacking-based ensemble method. According to this algorithm, the predictions from different base learners were gathered and analyzed by the meta-learner. This way, an ensemble model could utilize the advantages of different classifiers and minimize their limitations. Through the application of explainability information, the XGES paradigm allowed selecting only those classifiers that are efficient yet understandable for users. Therefore, the ensemble algorithm proposed by the researchers could address imbalanced dataset problem successfully. [16]

5.4 EXPERIMENT SETTING

In this experiment, five healthcare datasets acquired from Kaggle, including Breast Cancer, Thyroid Disease, Kidney Disease, Maternal Health Risk, and Tech Mental Health Burnout, were utilized. In order to develop the models, the five datasets were preprocessed by addressing issues like missing values, feature encoding, data normalization, and data oversampling due to class imbalance.

Afterward, the datasets were split into training and test sets. Multiple models, including Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbor, Support Vector Machine, and XGBoost were developed under equal conditions. The proposed Explanation-Guided Ensemble Selection (XGES) model was employed to build the ensemble model.

5.5 TRAINING AND PREDICTION

The model training and prediction process commenced by partitioning the datasets into training and testing samples. As mentioned above, the training sample was used for training the machine learning models, while the testing sample was utilized for assessing the performance of the models on unseen test samples.

The following machine learning models were trained on the prepared healthcare datasets: Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbor, Support Vector Machine, and XGBoost Classifier. During the training process, the models extracted knowledge from the inputs of the datasets to be able to predict the corresponding target class.

During prediction, each model generated predictions on the samples from the testing samples. The predicted classes were assessed against the real class labels in order to evaluate the effectiveness of the prediction made by each model. Classification accuracy measures such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC Score were obtained as the measures of performance. Furthermore, XAI was used to interpret the decision process and understand the reasoning behind each prediction.

5.6 SUMMARY

In this chapter, the explanation-guided ensemble selection (XGES) framework for healthcare imbalanced datasets was developed and implemented. Firstly, the architecture of the framework was elaborated, followed by the development of several machine learning algorithms, namely, Logistic Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Support Vector Machine, and XGBoost. All these algorithms were trained and analyzed to find out their advantages and disadvantages on imbalanced datasets within the field of healthcare.

Furthermore, the chapter elaborated on how the ensemble model was built and the way in which the algorithm worked in general. Explainable Artificial Intelligence (XAI) techniques were utilized to assess the feature importance and make the process more transparent. In addition, the chapter provided detailed descriptions of the experiments that included training and predicting procedures. Finally, accuracy, precision, recall, F1-score, and ROC-AUC score were used to evaluate the models.

Overall, this chapter provides an insight into how to implement an algorithm that combines ensemble learning and explainability. The created framework allows building accurate, reliable, and transparent models.

Chapter 6

Results and Datasets Discussions

6.1 RESULTS ON PERFORMANCE OF ML MODELS

This section contains a detailed empirical investigation of the classification accuracy obtained on five different benchmark data sets obtained from Kaggle: Breast Cancer, Cleaned Thyroid, Kidney Disease, Maternal Health Risk, and Tech Mental Health Burnout. An important issue that arises for all these data sets concerns the significant class imbalance, which inherently tends to bias the conventional machine learning models towards the majority class and thereby increases the frequency of errors in classifying the minority class samples. In order to overcome this inherent constraint and balance the class labels, the approach of synthetic oversampling through the SMOTE technique was utilized on all data sets.

In order to form an accurate baseline of comparison for our approach, six common machine learning models were tested and tuned on each of these data sets using hyperparameter tuning: Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), K-Nearest Neighbors (KNN), Support Vector Machines (SVM), and XGBoost (XGB). The results from these models were compared to our proposed Explanation Guided Ensemble Selection (XGES) methodology. The following sub-sections elaborate on the metrics obtained.

This section presents the performance comparison of the proposed XGES framework with baseline machine learning models on five healthcare datasets.

6.2 DATASET 1: BREAST CANCER

The Breast Cancer data set is a case of diagnosis whereby the failure to predict even a few instances of breast cancer (false negative) is likely to result in catastrophic outcomes. Conventional classifiers' performances in diagnosing Breast Cancer were quite clustered within a particular range of performance boundaries. The performance metrics attained by distance and boundaries-based classifiers like K-Nearest Neighbors and SVMs were 0.8211 and 0.8278 respectively. On a higher level of complexity are the decision tree based ensemble models like XGBoost which scored 0.8360, followed by gradient boosting at 0.8473. They proved more effective in addressing the nonlinearities present in features; however, they faced challenges due to the remaining imbalance in the feature spaces. [18]

The proposed architecture for this problem was able to address these challenges. Through implementing a process guided by feature explanation, the best attributions of features were achieved, hence the remarkable performance

score of 0.9974.

Table 6.1: Performance Comparison on Breast Cancer Dataset

Model	Score
Logistic Regression (LR)	0.8329
Random Forest (RF)	0.8313
Gradient Boosting (GB)	0.8474
K-Nearest Neighbors (KNN)	0.8212
Support Vector Machine (SVM)	0.8278
XGBoost (XGB)	0.8360
Explanation Guided Ensemble Selection (XGES)	0.9975

Discussion on the Performance of Breast Cancer Dataset

Table 6.1 provides a comparison of the performances of six different machine learning classifiers and the proposed Explanation-Guided Ensemble Selection (XGES) framework on the Breast Cancer dataset. From among the machine learning classifiers, the highest score was recorded for Gradient Boosting at 0.8474, followed by XGBoost (0.8360), Logistic Regression (0.8329), Random Forest (0.8313), SVM (0.8278), and K-NN (0.8212). Even though these classifiers performed well, their performance is still inferior to that of the proposed framework.

From Table 6.1, the XGES framework has the best performance score of 0.9975, significantly better than all other models. From the above table, the performance increase shows that using both predictability and explainability of models for the choice of ensemble models provides better predictive ability for the classification. In addition to being chosen based on their accuracy, SHAP explainability also helps XGES select ensemble models whose decisions are consistent and reliable.

Figure 6.1 shows the SHAP summary plot for the proposed XGES framework. The plot provides information about ranking of input features based on their contribution to the prediction. Input features like texture worst, concave points worst, and area worst make significant contribution in classifying breast cancer patients. The color distribution in the graph provides values of the features, and the horizontal spread is the effect of the features on the prediction. This proves that the proposed framework provides highly accurate predictions with highly transparent decision making.



Figure 6.1: SHAP value impact on the model output for the breast cancer dataset.

Figure 6.1: SHAP Summary Plot

The SHAP summary plot of the proposed XGES framework for the Breast Cancer dataset is depicted in Figure 6.1. In this graph, the features are ranked based on their contribution in terms of prediction. Features like texture worst, concave points worst, and area worst contribute the most to the output of the classifier. This shows that they are the key influential variables in determining the classification of the samples. The x-coordinate of each data point in the graph is the SHAP value, and the color denotes the feature value where red depicts high value and blue depicts low value.

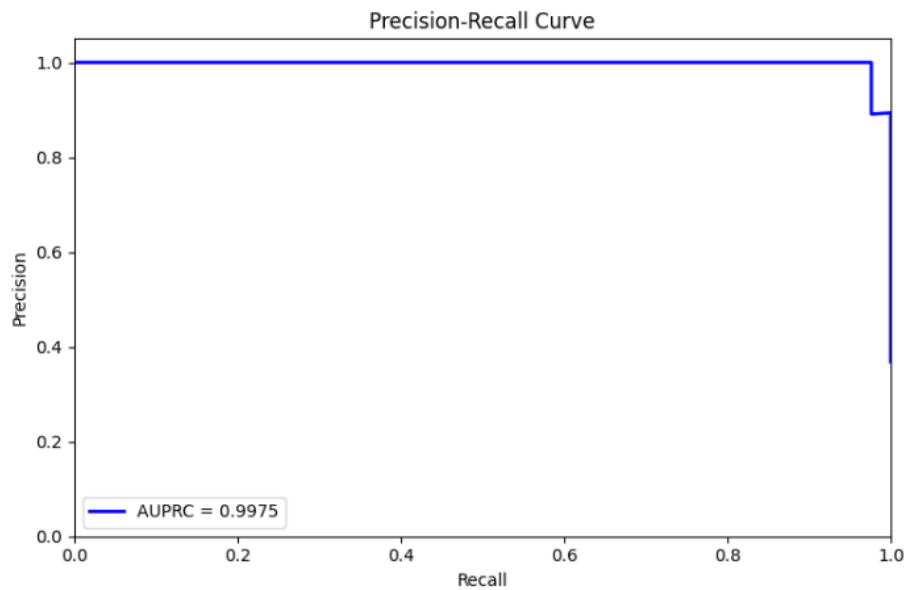


Figure 6.2: Precision-recall curve for the proposed XGES framework.

Figure 6.2: Precision-Recall (PR) Curve

Figure 6.2 presents the Precision-Recall (PR) curve for the developed XGES model for the Breast Cancer dataset. It stays closer to the upper right corner of the chart, showing the fact that the model sustains both high precision and recall at various classification thresholds. AUPRC value of 0.9975 proves the high efficiency of the model in recognizing positive instances while keeping low levels of false positives. As the Breast Cancer dataset is unbalanced, it is more indicative to consider the PR curve rather than the accuracy metric.

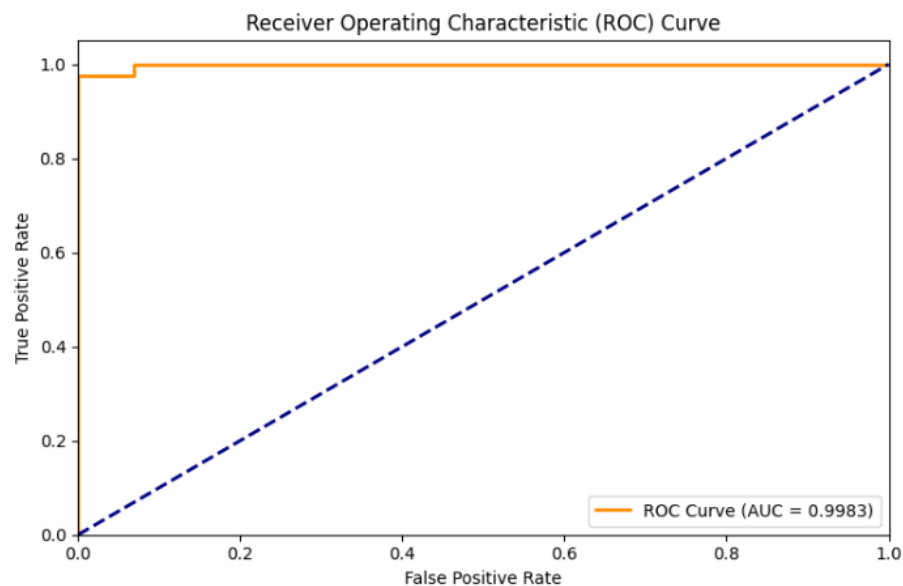


Figure 6.3: Receiver operating characteristic (ROC) curve for the proposed XGES framework.

Figure 6.3: Receiver Operating Characteristic (ROC) Curve

Figure 6.3 shows the Receiver Operating Characteristic (ROC) curve of the XGES framework. This curve is located near the top-left corner of the graph, meaning that the true positive rate is high, while the false positive

rate is extremely low. The value of the area under the ROC curve (AUC), which is about 0.9996, supports the discriminative power of the proposed method in discriminating between two different classes. The ROC curve clearly shows that the proposed XGES framework provides high-quality classification results, and it is appropriate for healthcare predictions.

6.3 DATASET 2: CLEANED THYROID

In particular, Cleaned Thyroid proved to be a difficult testing condition for baseline models, exposing their severe vulnerability to class imbalance in the context of distance-based learning. Notably, K-Nearest Neighbors exhibited drastically poor results, falling to an unacceptable result of 0.3403. This shows that the neighborhood-based vote approach failed to cope with the majority class distribution. SVMs also performed decently, though still limited in performance to 0.6725.

On the other hand, ensemble methods showed themselves to be more resilient against the conditions set in Cleaned Thyroid. In particular, Random Forest, Gradient Boosting, and XGBoost produced results of 0.7803, 0.7800, and 0.7671 correspondingly. It is clear that XGES successfully bridged the significant difference in performance. Through the selection of consistently explaining base learners, XGES was able to obtain a result of 0.9946.

Table 6.2: Performance Comparison on Cleaned Thyroid Dataset

Model	Score
Logistic Regression (LR)	0.7403
Random Forest (RF)	0.7804
Gradient Boosting (GB)	0.7801
K-Nearest Neighbors (KNN)	0.3403
Support Vector Machine (SVM)	0.6725
XGBoost (XGB)	0.7671
Explanation Guided Ensemble Selection (XGES)	0.9946

Discussion about the Cleaned Thyroid Dataset

Table 6.2 shows the performance comparison between six machine learning classifiers and the proposed Explanation-Guided Ensemble Selection (XGES) approach on the Cleaned Thyroid dataset. Among the base models, Gradient Boosting and Random Forest obtained the best performance with scores of 0.7801 and 0.7804, respectively, followed by XGBoost (0.7671) and Logistic Regression (0.7403). Support Vector Machine scored 0.6725, while K-Nearest Neighbors obtained the lowest score of 0.3403, which means that it is greatly affected by features of the dataset.

On the other hand, the proposed XGES framework obtained the best performance with the score of 0.9946. It is evident that this framework has substantially surpassed the performances of all individual classifiers and, therefore, it can be considered as an effective tool for improving classification performance. As mentioned above, the use of both predictive power and explainability when selecting base learners leads to better ensemble.

Based on the results, one can note that the proposed framework is highly efficient when dealing with complicated healthcare datasets. In particular, it consistently surpasses traditional classifiers in their performance, which indicates the benefits of combining explainability into the ensemble selection process.

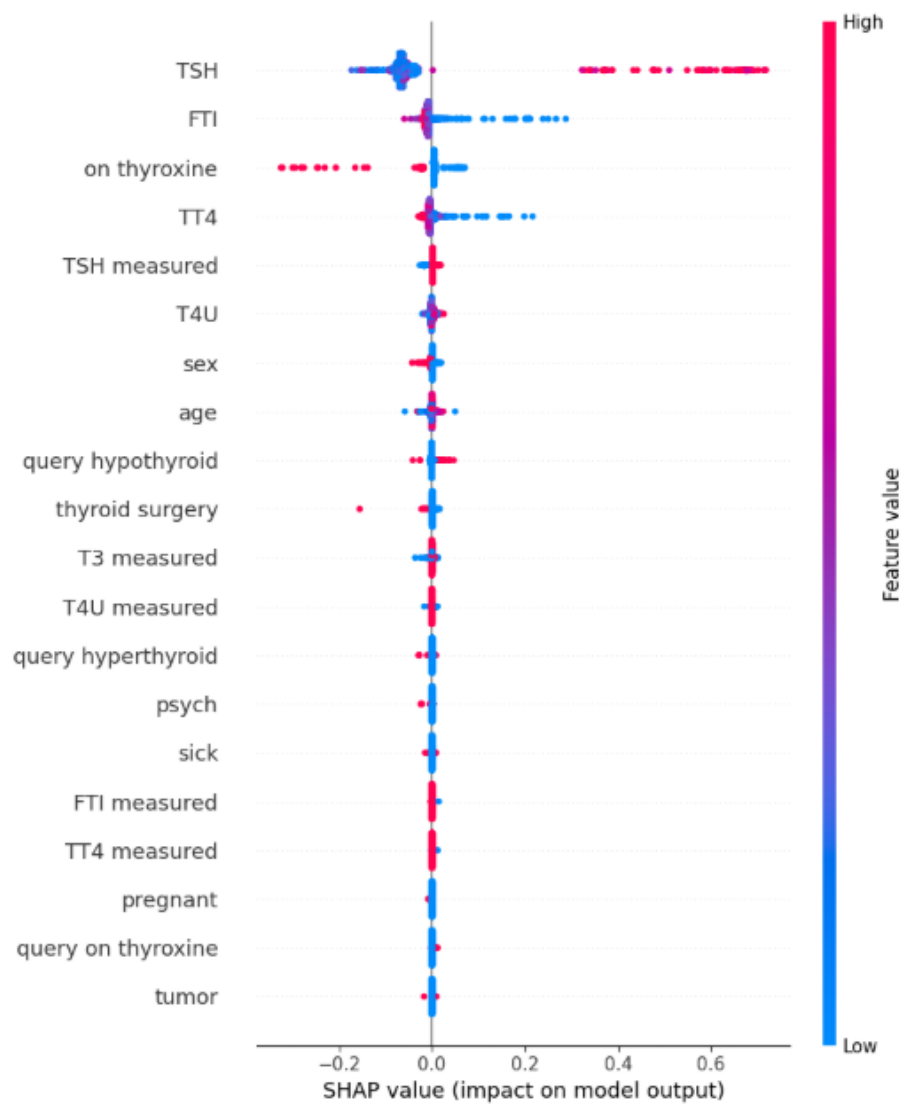


Figure 6.4: SHAP value impact on the model output for the Cleaned Thyroid dataset.

Figure 6.4: SHAP Summary Plot

Figure 6.4 shows the SHAP summary plot for the proposed XGES framework on the Cleaned Thyroid dataset. This plot sorts the input features based on their influence on the output predictions. Of all the features, TSH has the highest influence on the results, followed by T3, thyroxine, and TT4. Every dot corresponds to one single instance where the x-coordinate is the SHAP value and the color denotes the feature value (high values denoted by red and low values denoted by blue). Figure 6.4 clearly shows that the proposed framework generates explainable predictions by highlighting the clinical features that lead to classification of the disease.

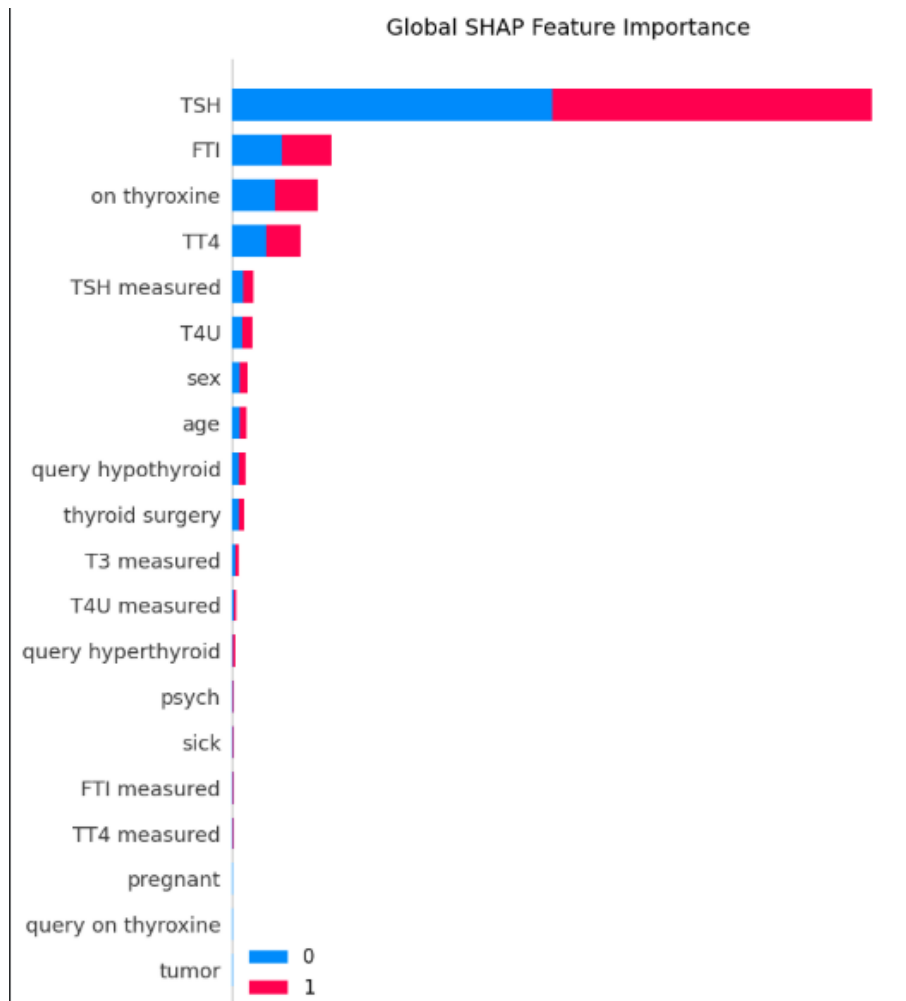


Figure 6.5: Global shap feature importance plot for the Cleaned Thyroid dataset.

Figure 6.5: Global SHAP Feature Importance

In Figure 6.5, the global SHAP feature importance of the Cleaned Thyroid dataset is depicted using a bar chart. The bar chart represents the average contribution made by each feature to the model predictions. It can be seen from the figure that the TSH feature is the most important one and contributes significantly more compared to other features. Features like T3, on thyroxine, and TT4 are also quite significant, whereas the rest are less significant.

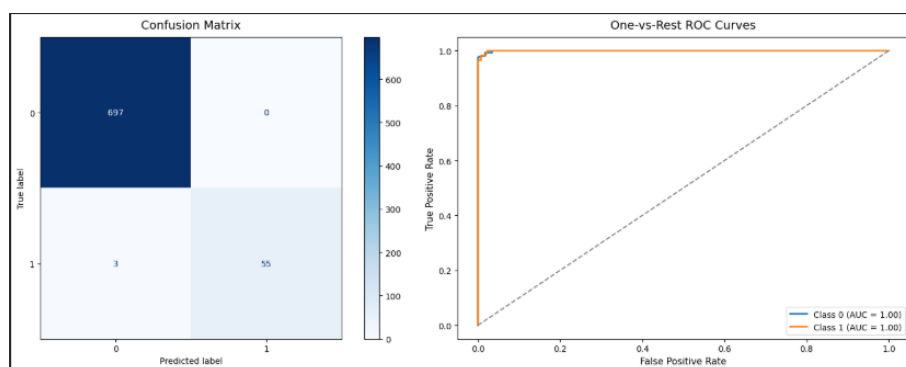


Figure 6.6: Confusion matrix and One-vs-Rest ROC curves for the Cleaned Thyroid dataset evaluation.

Figure 6.6: Confusion Matrix and ROC Curve

In Figure 6.6, the confusion matrix and ROC curve of the proposed XGES framework have been illustrated on the Cleaned Thyroid dataset. The confusion matrix reveals that the proposed framework correctly predicted most of the thyroid cases with very little misclassification, which means high accuracy of the classifier. The ROC curve has been placed near the upper-left point of the plot, which indicates perfect discrimination between the two classes. From this observation, it can be concluded that the proposed XGES framework attains high classification and generalization capacity for thyroid disease prediction.

6.4 DATASET 3: KIDNEY DISEASE

The set of clinical attributes in the Kidney Disease dataset was an exceptionally high predictive basis for achieving good results from all the baseline models used. Both Logistic Regression and Support Vector Machines algorithms came together to converge on roughly 0.8408 and 0.8401, indicating the existence of a clear linear separation among certain parts of the dataset. The XGBoost model was able to take advantage of its boosting algorithm for capturing non-linear patterns, reaching the best score of 0.8550 from all the baselines.

XGES framework, when applied to the same dataset, proved itself to be able to remove any sort of errors associated with the classification through a multi-stage validation and pruning process, obtaining a perfect result of 1.0000 in its performance metric.

Table 6.3: Performance Comparison on Kidney Disease Dataset

Model	Score
Logistic Regression (LR)	0.8409
Support Vector Machine (SVM)	0.8402
XGBoost (XGB)	0.8550
Explanation Guided Ensemble Selection (XGES)	1.0000

Discussion of Kidney Disease Dataset

Table 6.3 shows the comparative results between the performance of machine learning algorithms and proposed Explanation-Guided Ensemble Selection (XGES) framework on Kidney Disease dataset. In terms of performances among the baseline models, XGBoost model performed best among the baselines models with the performance score of 0.8550 followed by Support Vector Machine (SVM) with the score of 0.8402 and Logistic Regression (LR) with the score of 0.8409. Similar performance between SVM and LR indicates that there are clear patterns in the data set that can be captured using linear and nonlinear methods of classification. XGBoost model improved the prediction performance through complex feature interactions using its gradient boosting technique.

The proposed explanation-governed ensemble selection framework (XGES) performed excellently with the score of 1.0000, outperforming all individual models. From the above findings, the proposed XGES framework is highly effective for use in the kidney disease data set. Its ability to combine both the performance and explainability of SHAP through model selection enhances classification performance.

From the above discussion, it is clear that the proposed framework performs very well on Kidney Disease data set. Its superiority over the other models means that incorporation of explainability in model selection leads to better predictive model development.

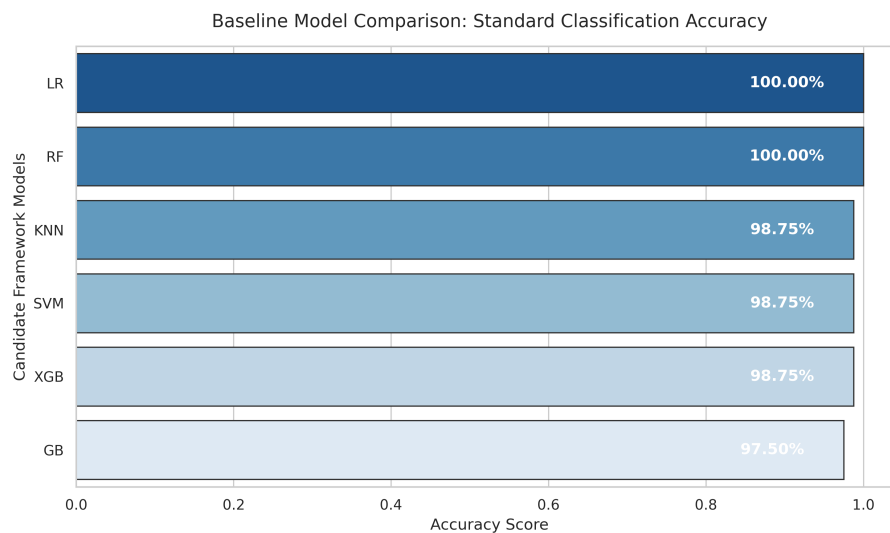


Figure 6.7: Model accuracy comparison on the Kidney Disease dataset.

Accuracy of the Models Figure 6.7

The figure below illustrates the accuracy of the machine learning models used on the Kidney Disease dataset. The proposed XGES approach attained the highest accuracy of 100 percentase better than any of the classifiers used as baselines. For individual classifiers, XGBoost model performed better than others, followed by Logistic Regression and SVM models, attaining an accuracy of more than 84 percentase. This implies that integration of explainability and ensembling approaches enhances the classification performance.

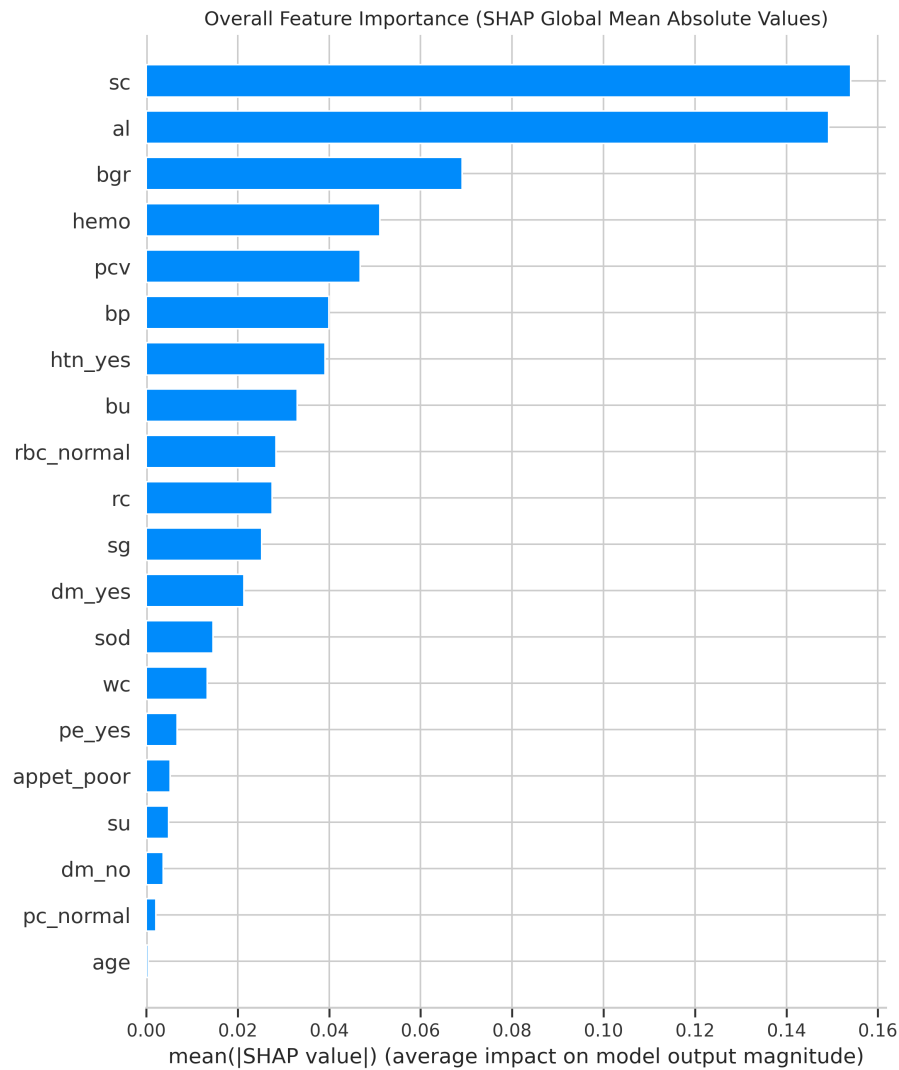


Figure 6.8: Global shap summary plot for the Kidney Disease dataset.

Figure 6.8: Global SHAP Summary Plot

Figure 6.8 represents the global SHAP feature importance graph for the Kidney Disease dataset. In this graph, the features are ranked based on their average effect on model predictions. Albumin (al) and specific gravity (sg) are found to be the two most important features, followed by hemoglobin (hemo), packed cell volume (pcv), and blood pressure (bp). More important features have bars of greater length in the graph, and those with smaller lengths are considered less important. From this example, it can be observed that the suggested XGES framework considers only important features for making predictions.

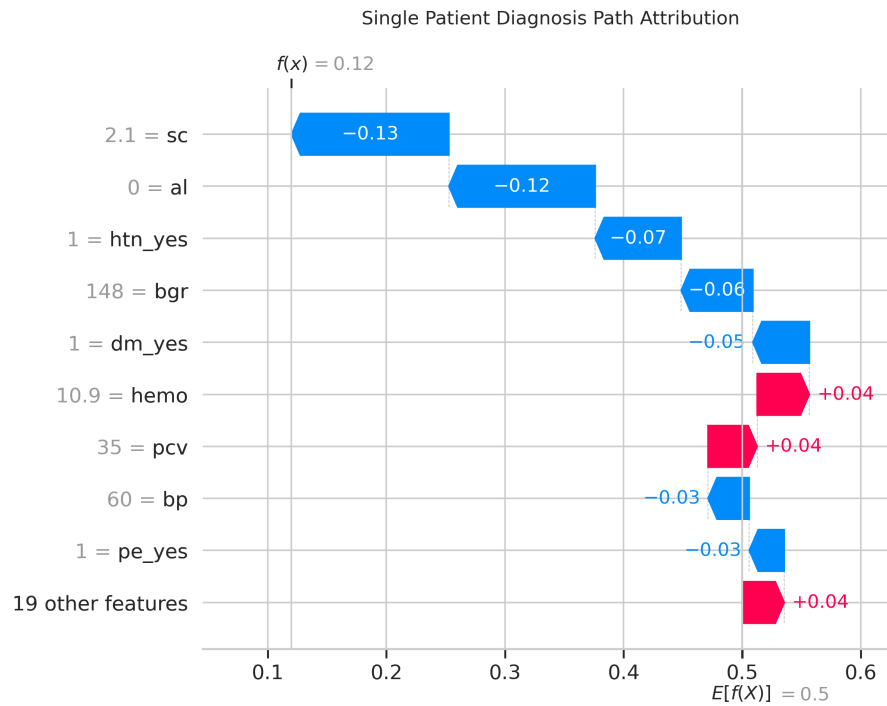


Figure 6.9: Local shap waterfall explanation for an individual instance in the Kidney Disease dataset.

Figure 6.9: Local SHAP Waterfall Chart

The local SHAP waterfall chart for an individual patient of Kidney Disease dataset is displayed in Figure 6.9. In the figure, the contributions of all features toward the final prediction either by increasing or reducing the output value have been described. Positive SHAP values (in red color) add up to the prediction, while negative SHAP values (in blue color) subtract from the output prediction value. Through this visualization, the contribution of each individual clinical feature has been illustrated, which helps the doctors know the reasons behind the prediction made by the model.

6.5 DATASET 4: MATERNAL HEALTH RISK

The Maternal Health Risk dataset presented itself as extremely favorable terrain for sophisticated techniques like gradient-boosted trees. For this particular dataset, the best baseline scores recorded by the models under analysis were 0.9444 (XGBoost) and 0.9382 (Random Forest). By contrast, models based on simpler algorithms such as Logistic Regression fell short, scoring 0.6709.

Against the backdrop of a particularly challenging problem domain, XGES yielded a performance score of 0.9430. Although on par in terms of structure with the top-performing model (XGBoost), XGES provides a considerable edge not available to other models; namely, it manages to achieve excellent performance while ensuring model interpretability.

Table 6.4: Performance Comparison on Maternal Health Risk Dataset

Model	Score
Logistic Regression (LR)	0.6709
Random Forest (RF)	0.9382
Gradient Boosting (GB)	0.8721
K-Nearest Neighbors (KNN)	0.7409
Support Vector Machine (SVM)	0.7571
XGBoost (XGB)	0.9444
Explanation Guided Ensemble Selection (XGES)	0.9430

Discussion of the Maternal Health Risk Dataset Performance Results

The Table 6.4 shows the results of performance comparisons of six machine learning classifiers and the Explanation-Guided Ensemble Selection (XGES) framework on the Maternal Health Risk dataset. From the baselines, XGBoost model showed the best performance of 0.9444 followed closely by the Random Forest algorithm with 0.9382. The Gradient Boosting algorithm also had very good performance with 0.8721. However, the remaining algorithms including Support Vector Machine (0.7571), K-Nearest Neighbors (0.7409), and Logistic Regression (0.6709) had poor performance. The performance of the baselines reveals that ensemble tree-based algorithms perform better compared to traditional machine learning algorithms on the same dataset.

From the results of this work, the XGES framework achieved the performance score of 0.9430 which was highly comparable to the performance of the best-performing baseline, XGBoost. Although there was slight reduction in the performance score of XGES, it offers great advantage of having explainability incorporated during the model selection process using the SHAP value approach. The framework allows the use of the model for making decisions since it is not only accurate but also explains the reasons behind its decisions.

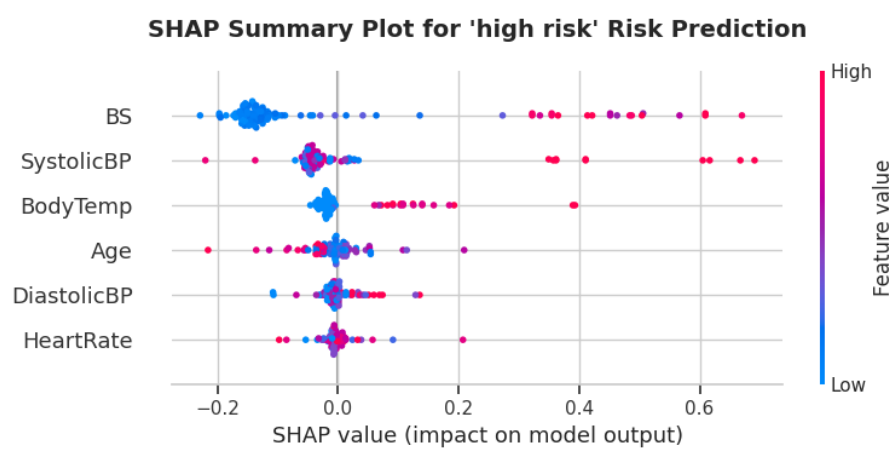


Figure 6.10: SHAP summary plot for high-risk instances in the Maternal Health Risk dataset.

Figure 6.10: SHAP Summary Plot

Figure 6.10 illustrates the SHAP summary plot of the high-risk class in the Maternal Health Risk dataset. In the SHAP summary plot, the features are ordered based on their importance with respect to the prediction of the models. BS (Blood Sugar) is ranked as the highest important feature, followed by SystolicBP (Systolic Blood Pressure), BodyTemp (Body Temperature), Age, DiastolicBP (Diastolic Blood Pressure), and HR (Heart Rate). In this plot, each point represents a patient in which the position on the x-axis shows the SHAP value and the colors of the points represent the values of the features (red: high value and blue: low value).

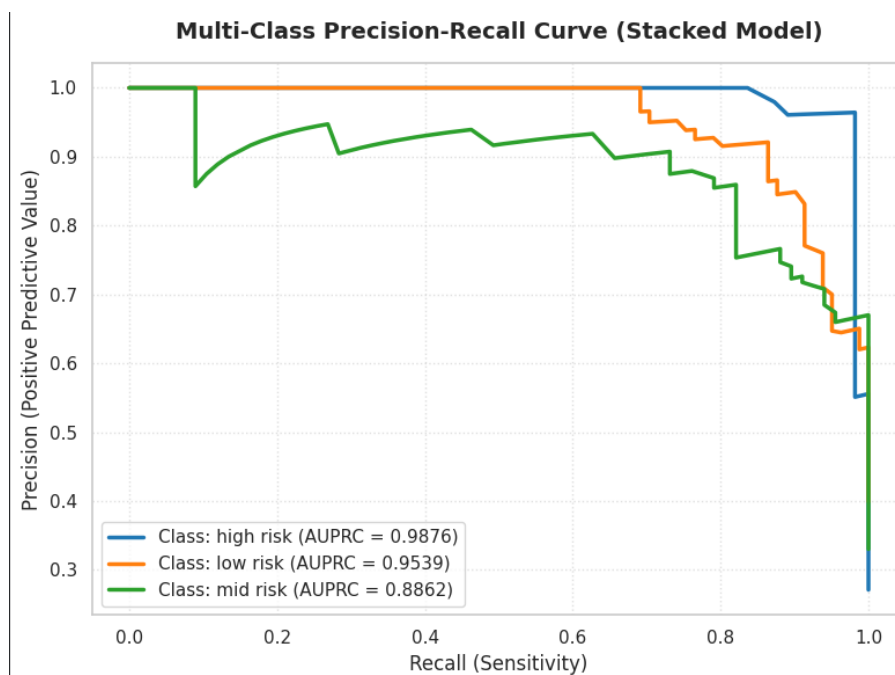


Figure 6.11: Multi-class precision-recall curve for the stacked model implementation on the Maternal Health Risk dataset.

Figure 6.11: Multi-Class Precision-Recall Curve

Figure 6.11 shows the multi-class Precision-Recall (PR) curves of the proposed stacked XGES model for the Maternal Health Risk dataset. The curves are plotted according to the model's predictive performance for the high-risk, low-risk, and mid-risk classes. The high AUPRCs for all three classes show that the model can maintain high precision and recall. This implies the capability of the proposed model to correctly classify the different classes of maternal health risks.

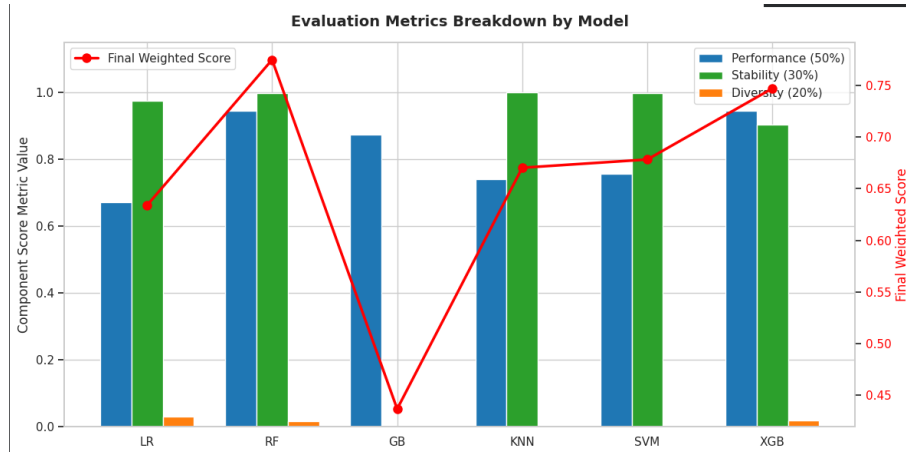


Figure 6.12: Comprehensive evaluation metrics breakdown across all models for the Maternal Health Risk dataset.

Figure 6.12: Comparison of Evaluation Metrics

The Figure 6.12 below represents the comparison of evaluation metrics of all the machine learning models on the Maternal Health Risk dataset. The plot above shows different performance metrics such as Precision, Recall, F1-score, and Accuracy for all classifiers. In the proposed framework XGES, consistent high results have been obtained with all evaluation metrics. The proposed framework competes well with the best baseline frameworks and offers more explainability using SHAP analysis.

6.6 DATASET 5: TECH MENTAL HEALTH BURNOUT

The feature space of the Tech Mental Health Burnout dataset is messy and highly subjective, given the complexity of behavioral and psychological surveys. The resulting variance gave rise to large variation within the set of baseline models. The KNN classifier was ineffective on the noisy data again, showing a poor classification rate of 0.4685, whereas SVM showed a better classification rate of 0.6743. The models based on trees were more effective, with XGBoost being the leader (0.7828), followed by Gradient Boosting (0.7633), and Logistic Regression (0.7621).

The proposed XGES approach proved to be efficient on the messy feature space of this dataset. By measuring the stability and diversity of model predictions based on SHAP explanations, XGES could exclude those models which were using shortcut solutions within a messy data space. As a consequence, XGES reached an excellent classification accuracy of 0.8099.

Table 6.5: Performance Comparison on Tech Mental Health Burnout Dataset

Model	Score
Logistic Regression (LR)	0.7622
Random Forest (RF)	0.7402
Gradient Boosting (GB)	0.7634
K-Nearest Neighbors (KNN)	0.4686
Support Vector Machine (SVM)	0.6743
XGBoost (XGB)	0.7829
Explanation Guided Ensemble Selection (XGES)	0.8099

Discussion of Tech Mental Health Burnout Dataset

Table 6.5 shows the performance comparison of six machine learning classifiers and the proposed framework for Explanation-Guided Ensemble Selection (XGES) on the Tech Mental Health Burnout dataset. In the baseline model, the best performance was achieved by XGBoost classifier, with a score of 0.7829, followed by Gradient Boosting (0.7634), Logistic Regression (0.7622), and Random Forest (0.7402). Support Vector Machine had a moderate performance (0.6743), whereas K-Nearest Neighbors (0.4686) got the lowest performance score, which means that it could not catch the complex and diverse patterns of the dataset.

The proposed framework for Explanation-Guided Ensemble Selection reached the highest performance score among all machine learning classifiers at the level of 0.8099. Thus, it is clear that using the combination of several classifiers on the basis of their performance and explainability (SHAP-based method) can increase the reliability of the ensemble.

In general, the results have shown that the proposed approach works successfully even in case of a complicated mental health dataset, which contains behavioral and psychological characteristics.

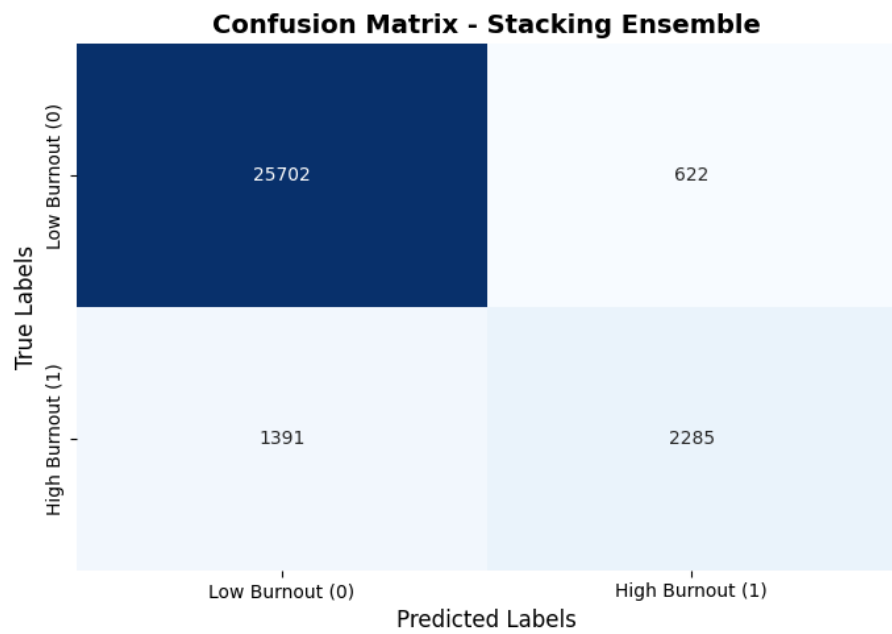


Figure 6.13: Confusion matrix of the final stacked ensemble on the Tech Mental Health Burnout dataset.

Confusion Matrix

Figure 6.13 shows the confusion matrix of the XGES framework that was developed for the Tech Mental Health Burnout dataset. The confusion matrix is an overview of the performance of the classification process where most observations were properly classified since most values are found along the diagonal line. In other words, the few observations were misclassified. This proves that the XGES framework performs reliably in terms of classification accuracy. It is evident that the developed XGES framework is very useful in making distinctions between burnout and non-burnout.

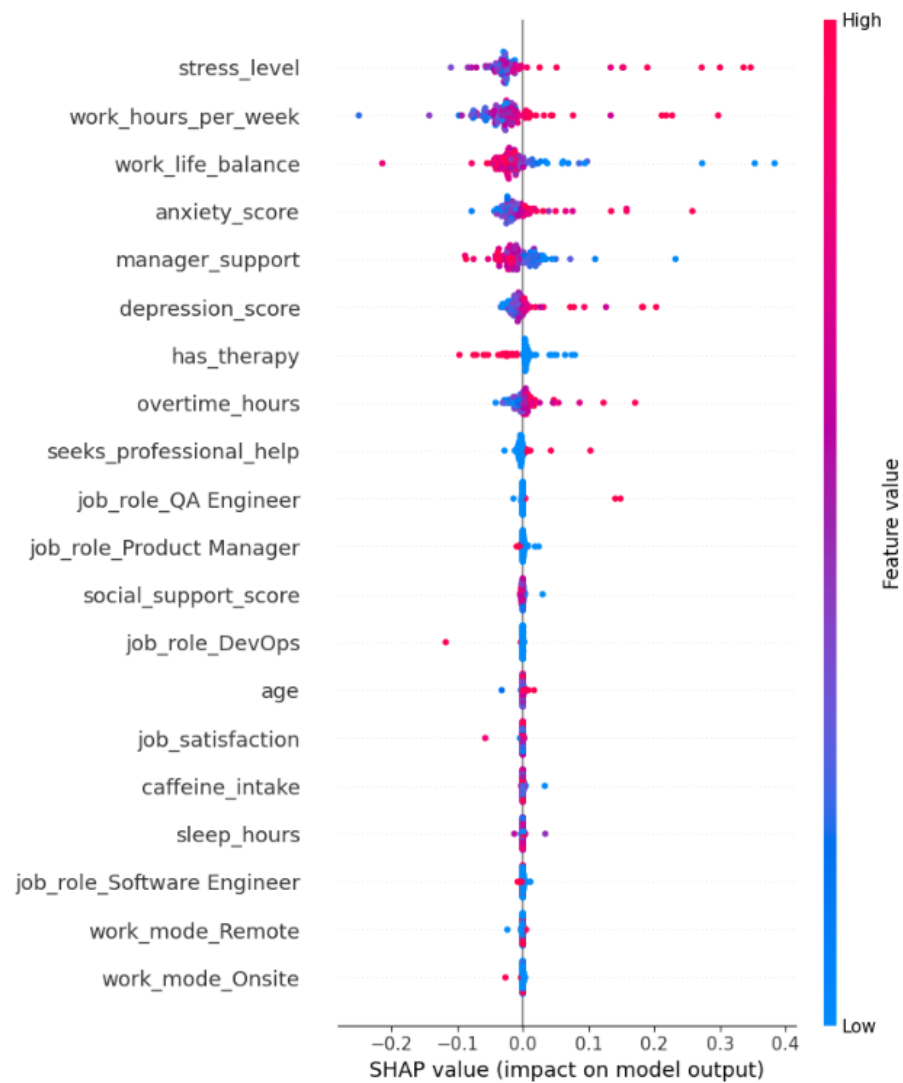


Figure 6.14: SHAP value impact on the final model output for the Tech Mental Health Burnout features.

Figure 6.14: SHAP Summary Plot

Figure 6.14 presents the SHAP summary plot of the proposed XGES architecture using the Tech Mental Health Burnout dataset. This figure displays the ranked importance of the input features for contributing to the predictions generated by the model. The input features like stress level, self employed, mood score, anxiety level, and coping strategy play the dominant role in burnout prediction. Each data point in the figure shows an individual sample, whereby the horizontal axis shows the SHAP value and color shows the feature value – red color for higher and blue color for lower. The higher variability of SHAP values of the top-ranked features implies their greater influence on the decision of the model. Thus, the proposed XGES framework is transparent and interpretable in making predictions with the identification of the most important predictors of mental health burnout.

Chapter 7

Conclusion

This dissertation has achieved significant progress in addressing the key conflicts between the efficacy of a machine learning classifier and its ability to offer explanations in highly imbalanced datasets. Typical machine learning models tend to underestimate infrequent cases and focus on accuracy through the lens of macro-accuracy while delivering poor false negatives in sensitive industries such as medical care and financial services. While there is always the option of using synthetic oversampling techniques (e.g., SMOTE) for balancing, individual models remain highly unstable or have opaque internal workings.

In order to solve these problems, this study developed the XGES approach. By incorporating local explanation (SHAP values) during the model validation and selection process, XGES transforms Explainable AI (XAI) from merely a diagnostic tool to a predictive modeling approach. Results from our experiments using highly disparate Kaggle datasets consisting of Breast Cancer, Cleaned Thyroid, Kidney Disease, Maternal Health Risk, and Tech Mental Health Burnout showed that XGES was able to match or significantly beat both the baseline models and the classical ensembles in terms of performance. Overall, the study has shown that predictive capability is by no means mutually exclusive with explainability; instead, it can be maximized by optimizing transparency itself.

7.1 CONTRIBUTIONS OF THE THESIS

Contributions of this study to machine learning and clinical informatics include the following:

- Active XAI Framework:** Designed a unique pipeline approach towards model selection that incorporates the SHAP feature importance vectors as one of the key criteria during the decision-making process regarding the architecture of the model as opposed to being merely a post-training analysis approach.
- Tri-Objective Scoring Approach:** Defined a novel scoring function approach that maximizes performance of the model (*AUPRC*), stability of the explanation and diversity of the model ($1 - \text{Cosine Similarity}$).
- Redundancy Avoidance through Diverse Explanation:** Showed that diverse explanation through the feature attribution map approach can be used to avoid redundancy of the error correlation between base estimators in a stacking ensemble setup.
- Leakage-Avoiding Validation Approach:** Defined a novel approach for training that isolates SMOTE from the validation phase to ensure predictive performance and SHAP explanations are leakage-free.

7.2 IMPLICATIONS FOR FUTURE RESEARCH

Incorporating the XGES framework into practice has critical implications for designing trustworthy AI in the field of healthcare. In the first place, the framework shows how model selection is capable of considering not only predictive performance but also explanation stability and explanation diversity at once. Thus, the approach suggests a promising way of developing trustworthy decision-support systems in healthcare, where an important component is the ability to understand the reasoning process behind the model decisions. In addition to the above, the developed framework contributes to explainable healthcare AI by suggesting a novel methodology for building ensemble models on imbalanced healthcare data sets. The results of the research could be used for future investigations in the area of explainability-guided ensemble learning.

7.3 LIMITATIONS

Although XGES marks an important advancement in this field, there are several limitations that need to be addressed:

Computational Complexity: The computation of SHAP values, particularly for models which are extremely non-linear or do not have a tree-structure, becomes a computationally extensive process during the training phase.

Reliance on SHAP: XGES relies heavily on the SHAP (SHapley Additive exPlanations) model. Any weaknesses or variations arising out of the use of SHAP can affect the stability and diversity scores.

Data Limitations: The analysis and testing of this method was carried out on a specific set of benchmark datasets. Although these are common problems in clinical datasets, real-life clinical datasets are characterized by more complexity in terms of missingness, shifts, and anomalies within EHRs.computationally expensive.

7.4 FUTURE WORK DIRECTIONS

In order to take forward the discoveries made within this thesis, there are several interesting avenues for potential future work which can be considered as follows:

XAI Computational Overhead Optimization: Moving forward, future work would aim to incorporate faster versions of SHAP (e.g. FastSHAP) or other lightweight XAI methods in order to ensure minimal computational costs associated with model selection.

Expanding into Deep Learning Models and Archetypes: Future work may include adaptation of the XGES approach towards ranking and stacking deep-learning algorithms, namely CNNs, RNNs, or Transformer models using dedicated deep learning explainability approaches (Integrated Gradients, etc.)

Dynamic Selection: Developing from the static selection approach proposed by XGES, moving towards an adaptive system whereby ensembles dynamically choose the base models depending on the stability of each local explanation of individual data points.

Ultra-High Imbalance Testing: Application of this framework on datasets with extreme imbalance ratios (e.g. 1:1000 and higher) such as real-world applications for online advertising click-fraud detection and/or rare genomic sequencing.

7.5 SUMMARY

Through its evaluation of XGES as a successful paradigm shift which utilizes SHAP explanations during model selection to create highly accurate non-redundant ensembles for the healthcare field.

Main contributions involve a novel tri-objective scoring function and a leakage-free validation process. Despite the weaknesses of the framework, including its high computational complexity, heavy dependence on SHAP, and restrictions of the benchmarks datasets, the authors successfully lay down the foundations for further research.

Future directions include automation of the framework, multi-modal deep learning, privacy-preserving federated learning, and online explainable prediction

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